

FINITE-REFERENCE ERROR BOUNDS FOR RECURRENT EVALUATORS IN POLICY IMPROVEMENT

Anonymous authors

Paper under double-blind review

ABSTRACT

We study fixed-snapshot error bounds for a finite-depth recurrent value evaluator inside policy improvement. On measurable state spaces, including continuous persistent latents, a finite reference depth $m > n$ controls value error through the depth discrepancy $\|U_n - U_m\|$ and a Bellman residual at depth m ; a recurrent path bounds the first term without requiring a recurrent latent fixed point. We compose this certificate with occupancy-aware conservative policy improvement (CPI): exact probability mixing gives the denominator $1 - \gamma + \gamma\alpha$, exact or approximate centering enters through a signed occupancy average, and policy overlap tightens value-to-advantage error. A stagewise finite-horizon result uses advantage spans, while a first-mismatch coupling gives a deployment denominator $1 - \gamma + \gamma\delta_{\text{dep}}$. Persistent latents use the clock-complete state (x, y, z, h) and one shared frozen recurrent transition map. These are conditional, fixed-MDP, fixed-policy-pair, fixed-snapshot results. Empirical evaluation is deferred.

1 INTRODUCTION

Tiny Recursive Models (TRMs) (Jolicoeur-Martineau, 2025) obtain predictions through iterative latent refinement. In checker-guided tasks such as Sudoku, the same recurrent computation can serve as an internal evaluator for candidate edits to a partial plan. A finite unroll can be useful while remaining transient: there is no general reason for an arbitrary TRM to be designed around convergence to a latent fixed point. Standard approximate-policy-iteration and conservative-policy-improvement (CPI) analyses (Bertsekas & Tsitsiklis, 1996; Kakade & Langford, 2002) place evaluator error in a black-box term and therefore do not distinguish a finite recurrent path from a Bellman residual at a chosen reference depth.

We analyze Unrolled Policy Iteration with TRMs (UPI-TRM), which retains the inner recurrence, uses a scalar value head to score candidate plan edits, and places environment transitions only in the outer checker-guided MDP. The analysis fixes the MDP, recurrence, value head, initialization, and policy pair. It is not an end-to-end theorem for SGD, TD learning, BPTT, or distillation. Euclidean projection can supply forward invariance, and exact baseline summation can supply the statewise centering needed by the CPI step, but these are conditions on a frozen snapshot rather than guarantees about how training reaches it.

Our primary result (Theorem 6.1) places this finite-reference argument on $B_b(\mathcal{C})$, so it covers measurable augmented states with $z \in \mathbb{R}^d$. It compares $U_n(s) = V_\psi(z^{(n)}(s), x, y)$ with any deeper finite reference U_m , bounding $\|U_n - V^\pi\|$ by $\|U_n - U_m\|$ plus the Bellman residual of U_m . No recurrent convergence is assumed, although the Bellman argument still uses the value-function fixed point V^π . Theorem 6.7 then composes the recurrent certificate with exact-mixture occupancy and advantage-span control. Theorems F.2 and F.5 give the stagewise and deployment counterparts. These are conditional fixed-snapshot results, not training guarantees.

We isolate the mathematical contribution and report no learned-task result, numerical theorem check, ablation, or method comparison. Empirical evaluation is deferred.

Scope. Write h for the remaining edit budget. Episodic latents are Markov on $s = (x, y, h)$. At a fixed snapshot, persistent latents are Markov on the clock-complete state $\bar{s} = (x, y, z, h)$. Their direct residual certificate needs no latent fixed point or slow drift; slow drift is needed only for the stronger

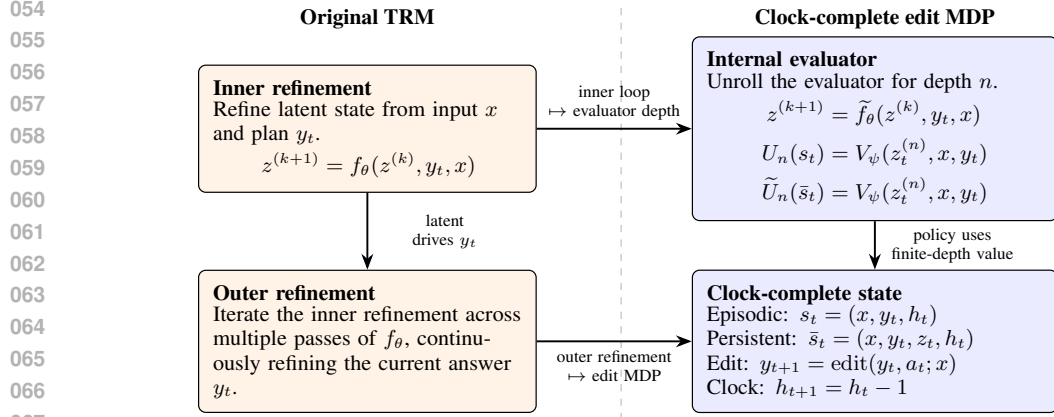


Figure 1: TRM-to-MDP bridge used throughout the paper. Left: the original TRM’s inner latent-refinement loop nested inside its outer continuous refinement of the current answer y_t (Jolicoeur-Martineau, 2025). Right: we swap TRM’s continuous outer refinement for a discrete plan-edit loop under checker feedback and treat the inner loop as an internal evaluator unrolled for depth n . Here $\tilde{f}_\theta$ is f_θ followed by the declared Euclidean projection or identity operator. The episodic Markov state is (x, y_t, h_t) ; carrying the latent requires the augmented state (x, y_t, z_t, h_t) . The checker maps to reward, and h_t records the remaining edit budget.

comparison with a memoryless fixed-point reference. The paper makes no learned-task performance claim.

Contributions.

- A measurable-space finite-reference certificate that does not require a recurrent latent fixed point, together with exact recurrent-path and finite-horizon specializations (Theorems 6.1 and F.1).
- A domain-specific composition of signed centering error, exact-mixture occupancy, advantage spans, and policy overlap with the recurrent evaluator certificate (Theorem 6.7 and Corollary 6.4).
- A direct certificate for persistent state on (x, y, z, h) , a finite-path latent comparison, and a sharp first-mismatch deployment composition (Corollary B.2, Lemma B.4, and Theorem F.5).

2 RELATED WORK

Recursive latent reasoning models. TRMs (Jolicoeur-Martineau, 2025) and Hierarchical Reasoning Models (Wang et al., 2025a) use iterative latent refinement for structured reasoning; follow-up work studies training dynamics and test-time adaptation (Qasim & Zhang, 2025; McGovern, 2025). We keep TRM’s inner latent refinement but swap its supervised outer loop for a discrete plan-edit loop under checker feedback, and analyze how finite inner evaluation affects policy improvement. The contraction assumption is DEQ-adjacent (Bai et al., 2019) only in providing a regime with bounded truncation bias.

Policy-improvement analyses. CPI, TRPO, and safe/monotone policy improvement (Kakade & Langford, 2002; Schulman et al., 2015; Pirodda et al., 2013; Metelli et al., 2021; Eshwar et al., 2025) provide policy-shift and occupancy-control templates. Function-approximation CPI (Metelli et al., 2021; Eshwar et al., 2025) is the closest formal neighbor for evaluator error; it relates a learned V_ψ to V^π through additional regression, sampling, or optimization assumptions. Our result is orthogonal: conditional on sup-norm residual and centering control, it exposes the finite recurrent path between depths n and m . A contractive recurrence turns this path into an L_z^n specialization, but contraction is not required by the primary result. Empirical RLVR work on LLM reasoning (Wang et al., 2025b; Su et al., 2025; DeepSeek-AI, 2025) uses verifiable rewards in a similar spirit but does not analyze approximation error. A quantitative join with FA-CPI on V_ψ would need coverage and uniform-convergence assumptions we leave out.

Bellman contraction and residual bounds, Banach fixed-point estimates, occupancy/span policy-improvement bounds, and coupling perturbation arguments are established tools. We do not claim these primitives in isolation. The contribution is their domain-specific synthesis for a recurrent evaluator: finite depth versus a deeper finite reference, measurable persistent-state Markovization, exact centering and probability mixing, and explicit composition with deployment discrepancy.

Planning and structured-decision neighbors. Sequence-model RL ranges from architecture-level planning (Tamar et al., 2016) and inference-time search (Janner et al., 2021) to pure sequence modeling (Chen et al., 2021; Laskin et al., 2023); structured prediction and Sudoku/constraint RL (Daumé III et al., 2009; Ross et al., 2011; Chang et al., 2015; Mehta, 2021; Klein, 2022) are closer to our plan-space MDP framing. The TRM inner unroll differs from both: it is architecture-internal computation within one state evaluation.

3 BACKGROUND AND SETUP

The original TRM, given input x and plan y_t , runs an inner loop that updates a latent $z^{(k+1)} = f_\theta(z^{(k)}, y_t, x)$. After n steps it reads out a refined prediction from a task head trained on checker-labeled puzzles (Jolicoeur-Martineau, 2025). UPI-TRM keeps f_θ but replaces the task head with two pieces. The first is a declared recurrent projection mode. Either $R > 0$ and $P_{\text{rec}} = \Pi_R$ is Euclidean projection onto the radius- R ball (Eq. 54), or projection is disabled and $P_{\text{rec}} = \text{Id}_{\mathcal{Z}}$. Set $\tilde{f}_\theta := P_{\text{rec}} \circ f_\theta$, $z^{(k+1)} = \tilde{f}_\theta(z^{(k)}, y_t, x)$. The convention $R = 0$ is not used to disable projection: Π_0 collapses every latent to zero. In projection-enabled mode, Π_R can keep latents inside B_R and support the optional contractive specialization of Assumption 5.1. The second is a scalar value head V_ψ that produces the unrolled evaluator output $U_n(s_t) = V_\psi(z_t^{(n)}, x, y_t)$ with episodic state $s_t = (x, y_t, h_t)$, where h_t is the remaining edit budget, and scores candidate plans in the outer CPI loop. The implemented head consumes the latent, instance, and current plan, but not h_t directly. The clock remains part of the Markov state because it determines termination, rewards, successors, and Bellman targets.

In the plan-space MDP the external episodic state is $s_t = (x, y_t, h_t)$, the action is an edit a_t , and a nonterminal transition sets $y_{t+1} = \text{edit}(y_t, a_t; x)$ and $h_{t+1} = h_t - 1$; solving, a terminal STOP action, or exhausting the budget sends the process to s_{abs} . Budget exhaustion is treated as termination, not as a separately bootstrapped truncation. The implementation uses an exactly return-equivalent folded terminal representation: every terminal cause sets the same terminal mask and folds the discounted absorbing tail into the terminal reward. Concrete terminal successor tensors are serialization payload only. Target and diagnostic paths may evaluate those payload tensors, but a nonfinite-safe terminal selection sets their successor-value contribution to zero, so they are never bootstrapped as distinct states. A checker score $c(x, y)$ supplies reward, optionally via potential shaping (Ng et al., 1999) $r = r_0 + \gamma\Phi(s') - \Phi(s)$ with $\Phi = c$. The latent z is *not* part of the environment state in the episodic analysis; it is internal computation used to evaluate s_t . The resulting process is a discounted MDP $(\mathcal{S}, \mathcal{A}, P, r, \rho, \gamma)$ with $\gamma \in [0, 1)$ and $K \geq 1$. State and action spaces are measurable, policies are stochastic kernels, and the reward, transition, recurrent-map, and value-head measurability needed by each expectation is stated below. All theoretical statements fix one parameter snapshot: the MDP, recurrence, initialization, value head, current policy π , and candidate policy π_{cand} do not change while the displayed quantities are evaluated. Value results use the ordinary pointwise sup norm on $B_b(\mathcal{C})$, not an essential supremum under an implicit reference measure. Finite/countable formulas are identified when they use probability mass functions or ℓ_1 . Increasing n spends more computation inside one state evaluation but does not change the environment horizon or search depth, so the analysis tracks evaluator truncation error inside a fixed external MDP.

Episodic- z vs. persistent- z . In *episodic- z* , the latent is reinitialized from the current plan at each edit step, so $s = (x, y, h)$ is Markov. In *persistent- z* , the latent is carried across edits, matching the original TRM. With persistent latents the next-state distribution depends jointly on (x, y, z, h) , since the same plan paired with a different carried latent or remaining budget can produce a different policy output or termination event. Treating (x, y) as the state in this regime is a non-Markov approximation. Appendix B lifts the finite-depth residual certificate directly to $\bar{s} = (x, y, z, h)$; this step needs no latent fixed point or slow drift. A separate slow-drift assumption is required only for comparison with a memoryless fixed-point reference.

Algorithm 1 One UPI-TRM update on the clock-complete edit MDP

```

1: Initialize: sample  $x$ , initialize  $y_0$ , choose  $T \geq 1$ , set remaining budget  $h_0 = T$ , and initialize  $z_0$  for
   persistent- $z$ 
2: for  $t = 0, \dots, T - 1$  do
3:   Set  $s_t = (x, y_t, h_t)$ ; for persistent- $z$ , set  $\bar{s}_t = (x, y_t, z_t, h_t)$ 
4:   Latent input: episodic- $z$  uses  $z_t \leftarrow z_{\text{init}}(x, y_t)$ ; persistent- $z$  uses the carried pre-unroll  $z_t$ 
5:   Evaluator:  $z_t^{(n)} \leftarrow \tilde{f}_\theta^{\text{on}}(z_t, y_t, x)$ 
6:   Value:  $U_n(s_t) \leftarrow V_\psi(z_t^{(n)}, x, y_t)$  in episodic- $z$ ;  $\tilde{U}_n(\bar{s}_t) \leftarrow V_\psi(z_t^{(n)}, x, y_t)$  in persistent- $z$ 
7:   Action: sample  $a_t \sim \pi_\phi(\cdot \mid y_t, z_t^{(n)}, x)$  and apply the edit; observe  $r_t, y_{t+1}$ , and terminal flag  $d_t$ ; set
      $h_{t+1} = h_t - 1$ 
8:   if  $d_t$  or  $h_{t+1} = 0$  then
9:     Store  $(s_t, a_t, r_t, s_{\text{abs}})$  in episodic- $z$ , or  $(\bar{s}_t, a_t, r_t, s_{\text{abs}})$  in persistent- $z$ 
10:    break
11:   else
12:     Set  $s_{t+1} = (x, y_{t+1}, h_{t+1})$  and  $\bar{s}_{t+1} = (x, y_{t+1}, z_t^{(n)}, h_{t+1})$ 
13:     Store  $(s_t, a_t, r_t, s_{t+1})$  in episodic- $z$ , or  $(\bar{s}_t, a_t, r_t, \bar{s}_{t+1})$  in persistent- $z$ 
14:     For persistent- $z$ , carry the post-unroll latent:  $z_{t+1} \leftarrow z_t^{(n)}$ 
15:   end if
16: end for
17: Value update: form  $G^{(K)}$  and regress the mode-appropriate evaluator ( $U_n$  or  $\tilde{U}_n$ )
18: Advantage estimate: compute  $\hat{A}$  centered conditional on  $s_t$ , or on full  $\bar{s}_t$  in persistent- $z$ 
19: Policy update: take a policy-gradient step to obtain  $\pi_{\text{cand}}$ 
20: Conservative step:  $\pi_{\text{new}} \leftarrow (1 - \alpha)\pi_\phi + \alpha\pi_{\text{cand}}$ 
21: Optional distillation: distill  $\pi_{\text{new}}$  into a single policy network

```

Notation. Define

$$\mathcal{C} := \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}, \quad \mathcal{C}^\circ := \mathcal{C} \setminus \{s_{\text{abs}}\}.$$

On $\mathcal{C}^\circ$, let $z^{(j)}(s)$ be the latent after j applications of the frozen recurrent map from one common measurable initialization, and set $U_j(s) = V_\psi(z^{(j)}(s), x(s), y(s))$. Write $b := V^\pi(s_{\text{abs}}) = -C_{\text{max}}$. At the absorber define only the depth-independent scalar boundary $U_j(s_{\text{abs}}) = b$ from Appendix E. The architectural specialization uses $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$, but Bellman contraction does not require initialization to ignore h . Value-function sup norms are taken on $\mathcal{C}$, while latent paths, fixed-point latents, and anchor distances are defined only on $\mathcal{C}^\circ$; no absorbing latent is introduced. Only the optional contractive specialization defines $z^*(s)$ and $U_*(s) = V_\psi(z^*(s), x(s), y(s))$ on $\mathcal{C}^\circ$, with the same scalar boundary $U_*(s_{\text{abs}}) = b$. Appendix D separates the π -invariant Bellman domain, the one-deviation advantage domain, the repeated-mixture occupancy domain, and the recursively invariant deployment domain. We write $\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} := \sup_{s \in \mathcal{C}} |f(s)|$.

4 UPI-TRM ALGORITHM

Algorithm 1 alternates three phases: collect edit trajectories, fit U_n against a K -step bootstrap target, and take a conservative mixture step. The exact-mixture CPI guarantee applies to the explicit mixture. A distilled policy is a different deployment policy and is covered only after a uniform TV or KL discrepancy from that mixture is established.

Every environment terminal flag, including a solved state, terminal STOP, or any other terminal cause, ends collection immediately after its terminal transition is stored. Budget exhaustion does the same. Only the nonterminal branch constructs a successor or carries a post-unroll latent.

The simplest exactly centered episodic and augmented-state corollaries of Theorem 6.7 require exact statewise baseline evaluation and the explicit mixture step. In the persistent case, centering and the mixture identity are conditional on the full augmented state. The master theorem also permits a signed centering defect. A clipped surrogate, trust-region update, parameter interpolation, or distilled approximation is a different policy: the exact-mixture occupancy term does not apply to it directly, but Corollary F.8 covers it when a uniform TV or KL discrepancy from the explicit mixture is established on the deployment domain.

Persistent- z is handled in Appendix B. In the projection-enabled mode, the initialization condition in Appendix J gives $C_z \leq 2R$. In a finite discrete-action implementation that computes the exactly centered baseline by explicit summation and uses one depth- n evaluator unroll for each action term, the evaluator portion costs $O(T|\mathcal{A}|n)$ recurrent-map applications: T states per trajectory, $|\mathcal{A}|$ action terms per state, and depth n per term. For general action spaces, exact kernel integration has implementation-dependent cost.

5 BELLMAN RESIDUALS AND AN OPTIONAL CONTRACTIVE REGIME

We train U_n by K -step bootstrapped regression with loss $\mathcal{L}_{\text{val}} = \mathbb{E}_{s_0 \sim \mu}[(V_\psi(z^{(n)}(s_0), x_0, y_0) - \text{sg}(G^{(K)}))^2]$, where $G^{(K)} = \sum_{k < K} \gamma^k r_k + \gamma^K \bar{V}(s_K)$ is the K -step bootstrap target. Conditional on s_0 , the target trajectory uses the frozen current policy π ; μ selects starting states and does not replace that conditional path law. The K -step Bellman operator $\mathcal{T}_K^\pi$ is the standard γ^K -contraction with fixed point V^π (Sutton & Barto, 2018; Puterman, 1994). Proposition 5.2 separates the target-network population residual from the self-bootstrap Bellman residual used by the certificate, and Corollary 6.2 propagates that distinction into the value and CPI bounds.

The primary finite-reference-depth result in Section 6 uses Bellman contraction but does not assume that the recurrent computation converges. It assumes sup-norm residual control on the relevant closure; we do not prove a general $L^2(\mu) \rightarrow \|\cdot\|_\infty$ conversion. A separate, optional assumption below makes a latent fixed-point specialization available. Bellman contraction is operator-level; inner-evaluator contraction of $z \mapsto \tilde{f}_\theta(z, y, x)$ controls a recurrent path within one frozen forward pass and by itself gives no TD- or training-convergence guarantee.

Assumption 5.1 (Domain-relative forward-invariant contraction). Fix a declared nonabsorbing state domain $\mathcal{D}^\circ$. The latent space $\mathcal{Z}$ is a complete normed space (in our implementation, $\mathbb{R}^d$ with the Euclidean norm). For $s \in \mathcal{D}^\circ$, set

$$T_s(z) := \tilde{f}_\theta(z, y(s), x(s)).$$

There exist a closed set $\mathcal{Z}_{\text{inv}} \subseteq \mathcal{Z}$ and $L_z \in [0, 1)$ such that, uniformly for every $s \in \mathcal{D}^\circ$,

$$T_s(\mathcal{Z}_{\text{inv}}) \subseteq \mathcal{Z}_{\text{inv}}, \quad \|T_s(z) - T_s(z')\| \leq L_z \|z - z'\| \quad (z, z' \in \mathcal{Z}_{\text{inv}}).$$

Every initialization used by the result satisfies $z_{\text{init}}(x(s), y(s)) \in \mathcal{Z}_{\text{inv}}$ for every $s \in \mathcal{D}^\circ$.

For the episodic finite-depth and projection-aware specializations, we invoke Assumption 5.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and additionally assume

$$C_z := \sup_{s \in \mathcal{C}^\circ} \|T_s(z_{\text{init}}(x(s), y(s))) - z_{\text{init}}(x(s), y(s))\| < \infty.$$

The persistent slow-drift specialization declares its augmented domain in Appendix B.2. A genuinely global contraction and initialization condition over every input pair is sufficient for any of these invocations, but is not implied by a closure-restricted premise.

We use $L_z^0 = 1$ for every $L_z \in [0, 1)$. When $L_z = 0$, we use $L_z^n = 0$ for every integer $n \geq 1$.

Proposition 5.2 (Target-network population residual bridge). *Let $\mathcal{E}$ be a measurable P_π -invariant state domain. Assume $r_K^\pi \in B_b(\mathcal{E})$ and $U_q, \bar{V} \in B_b(\mathcal{E})$ for a finite depth $q \geq 0$, and suppose the conditional K -step target trajectory follows the frozen policy π . Define*

$$\varepsilon_{\text{targ},q} := \|U_q - (r_K^\pi + \gamma^K P_\pi^K \bar{V})\|_{\infty, \mathcal{E}}. \quad (1)$$

Retain the propagated target lag by defining

$$R_{\text{targ},q}^P := \varepsilon_{\text{targ},q} + \gamma^K \|P_\pi^K (\bar{V} - U_q)\|_{\infty, \mathcal{E}}, \quad (2)$$

$$R_{\text{targ},q}^{\text{sup}} := \varepsilon_{\text{targ},q} + \gamma^K \|\bar{V} - U_q\|_{\infty, \mathcal{E}}. \quad (3)$$

Then the self-bootstrap residual satisfies

$$\|U_q - \mathcal{T}_K^\pi U_q\|_{\infty, \mathcal{E}} \leq R_{\text{targ},q}^P \leq R_{\text{targ},q}^{\text{sup}}. \quad (4)$$

For the formal absorbing representation, require $U_q(s_{\text{abs}}) = \bar{V}(s_{\text{abs}}) = b$ and use the same absorbing reward and boundary in both population operators. The equivalent folded implementation may instead fold γb into the terminal-transition reward and mask the terminal bootstrap to zero, but it must not add the absorbing tail again.

Proof. Add and subtract the target-network population backup:

$$U_q - \mathcal{T}_K^\pi U_q = [U_q - (r_K^\pi + \gamma^K P_\pi^K \bar{V})] + \gamma^K P_\pi^K (\bar{V} - U_q).$$

The triangle inequality gives the first bound in Eq. (4). Invariance makes P_π^K a Markov operator on $B_b(\mathcal{E})$, so it is nonexpansive in the pointwise sup norm and gives the second bound. $\square$

The empirical regression loss is not $\varepsilon_{\text{targ},q}$: even a population mean-square regression loss controls an $L^2(\mu)$ conditional residual, not either displayed sup norm. A finite-batch maximum establishes neither that conditional sup norm nor the propagated lag $\|P_\pi^K(\bar{V} - U_q)\|_{\infty, \mathcal{E}}$; it also does not establish the fallback $\|\bar{V} - U_q\|_{\infty, \mathcal{E}}$. The certificate remains conditional on independently establishing these uniform quantities. The fixed-target population backup is not the self-bootstrap Bellman operator; the quantities in Eqs. (2)–(3) only upper-bound its residual.

The implementation offers operator-norm clamping and per-layer scaling as contraction-oriented interventions; when projection is enabled, Π_R provides forward invariance. The Euclidean projection is nonexpansive, so composition with it does not increase a proved Lipschitz upper bound. It supplies strict contraction only when the final modulus proved uniformly on the declared domain is strictly below one. The empirical label *clamping enabled* does not imply contraction. Any finite-difference quantity is a local proxy and does not certify the uniform domain-relative modulus below one required by Assumption 5.1.

6 FINITE-DEPTH EVALUATION ERROR DECOMPOSITION AND CPI PENALTY

We connect finite evaluator depth to CPI. The CPI shell is standard (Kakade & Langford, 2002); the TRM-specific content is a finite-reference-depth error decomposition that does not require a recurrent fixed point. We focus on episodic- z and defer persistent- z to Appendix B.

6.1 VALUE ERROR BOUNDS (EPISODIC- z)

K -step Bellman contraction gives, for any bounded measurable V on the forward-invariant closure,

$$\|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{1}{1 - \gamma^K} \|V - \mathcal{T}_K^\pi V\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}, \quad (5)$$

and therefore permits any finite unroll depth to serve as the reference.

Theorem 6.1 (Measurable finite-reference-depth value decomposition). *Let $\mathcal{C}$ be a measurable state set with its trace σ -algebra, let P_π be a Markov kernel satisfying $P_\pi(\mathcal{C} \mid s) = 1$ for every $s \in \mathcal{C}$, and fix $\gamma \in [0, 1)$ and an integer $K \geq 1$. If an initial law is used, assume $\rho(\mathcal{C}) = 1$. Assume the one-step reward is uniformly bounded and its fixed-policy conditional mean $r^\pi(s) := \mathbb{E}_s^\pi[r_0]$ is measurable on $\mathcal{C}$. Then the K -step expected reward*

$$r_K^\pi(s) := \mathbb{E}_s^\pi \left[\sum_{i=0}^{K-1} \gamma^i r_i \right] \quad (6)$$

is bounded and measurable on $\mathcal{C}$. On the Banach space $B_b(\mathcal{C})$ of bounded measurable functions with the ordinary pointwise sup norm, define

$$(\mathcal{T}_K^\pi V)(s) := r_K^\pi(s) + \gamma^K (P_\pi^K V)(s). \quad (7)$$

Then $\mathcal{T}_K^\pi : B_b(\mathcal{C}) \rightarrow B_b(\mathcal{C})$ is a γ^K -contraction and has a unique fixed point V^π . For arbitrary endpoint functions $U_n, U_m \in B_b(\mathcal{C})$ and finite integers $0 \leq n < m$,

$$\|U_n - V^\pi\|_\infty \leq \underbrace{\|U_n - U_m\|_\infty}_{\text{finite-depth discrepancy}} + \underbrace{\frac{\|U_m - \mathcal{T}_K^\pi U_m\|_\infty}{1 - \gamma^K}}_{\text{reference-depth residual}}. \quad (8)$$

For the recurrent specialization, take $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ and $\mathcal{C}^\circ = \mathcal{C} \setminus \{s_{\text{abs}}\}$. On $\mathcal{C}^\circ$, fix a jointly measurable recurrent map $(s, z) \mapsto T(s, z)$, a common measurable initialization into its declared latent

domain, and a latent norm, and define $z^{(j)}$ by iterating that map and $U_j(s) = V_\psi(z^{(j)}(s), x(s), y(s))$. At the absorber define only the common scalar boundary $U_j(s_{\text{abs}}) = b$, and assume the two endpoint functions lie in $B_b(\mathcal{C})$. If the measurable scalar head is L_V -Lipschitz in its latent argument uniformly while (x, y) is held fixed, then

$$\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V \sup_{s \in \mathcal{C}^\circ} \|z^{(n)}(s) - z^{(m)}(s)\| \quad (9)$$

$$\leq L_V \sup_{s \in \mathcal{C}^\circ} \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\| \quad (10)$$

$$\leq L_V \sum_{j=n}^{m-1} \sup_{s \in \mathcal{C}^\circ} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \quad (11)$$

If $\mathcal{C}^\circ$ is empty, the nonnegative latent suprema in this chain are defined to be zero.

The Bellman and residual steps are standard measurable-space dynamic programming. Their role here is a domain-specific synthesis: they expose a finite recurrent reference m whose error is certifiable when the displayed uniform discrepancy and residual are available. The recurrence may perform useful transient computation and need not converge before, at, or after m . Only the endpoints U_n, U_m are needed for the abstract residual inequality; the common recurrence and initialization are needed for the path interpretation. The same frozen policy π appears in V^π and $\mathcal{T}_K^\pi$, and the latter bootstraps from U_m itself.

For the optional contractive specialization, invoke Assumption 5.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the finite constant C_z above. For each represented $s \in \mathcal{C}^\circ$, write $z^*(s)$ for the unique fixed point of T_s . The joint measurability and measurable initialization just imposed make $s \mapsto z^*(s)$ measurable by Proposition A.2. Invoke Assumption A.3 on $\mathcal{D}^\circ = \mathcal{C}^\circ$ with $\mathcal{Z}_{\text{bd}} = \mathcal{Z}_{\text{inv}}$; together with the separate finite absorbing boundary, this makes U_* bounded on the closure. Set $U_*(s_{\text{abs}}) = -C_{\text{max}}$ and define

$$\varepsilon_{\text{res}}^* \triangleq \|U_* - \mathcal{T}_K^\pi U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (12)$$

Under this domain-relative invocation of Assumption 5.1 and the Lipschitz-head condition, consecutive latent differences satisfy $\|z^{(j+1)} - z^{(j)}\| \leq L_z^j C_z$ uniformly. Hence Theorem 6.1 gives the finite- m specialization

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq L_V C_z \frac{L_z^n (1 - L_z^{m-n})}{1 - L_z} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (13)$$

Uniform convergence $U_m \rightarrow U_*$ and Bellman contraction imply convergence of the residual norm. Taking $m \rightarrow \infty$ recovers

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \underbrace{\frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{fixed-point residual}} + \underbrace{L_V \frac{L_z^n}{1 - L_z} C_z}_{\text{geometric path bound}}. \quad (14)$$

Thus the earlier fixed-point decomposition is a specialization obtained by geometrically bounding the recurrent path and sending the finite reference depth to infinity. It is not an assertion that arbitrary TRMs are designed to converge to a fixed point. Under $\|z_{\text{init}}\|_2 \leq R$ with $\mathcal{Z}_{\text{inv}} = B_R$, projection gives $C_z \leq 2R$ (Remark I.2).

A projection-active specialization makes the role of R explicit. When all pre-projection outputs represented by $s \in \mathcal{C}^\circ$ satisfy $\|f_\theta(z, y(s), x(s))\|_2 \geq \rho_R > R$ for $z \in B_R$, radial projection reduces the pre-projection Lipschitz upper bound by the factor R/ρ_R . The projected map is contractive on this domain only when $(R/\rho_R)L_z^{\text{pre}}(R) < 1$. Proposition I.3 makes this condition explicit and gives

$$L_z \leq \bar{L}_z^{\text{sat}}(R) := \frac{R}{\rho_R} L_z^{\text{pre}}(R), \quad (15)$$

and Proposition I.4 bounds the residual relative to the boundary-normalized zero-latent baseline

$$B_0(x, y) := V_\psi(0, x, y), \quad B_0(s_{\text{abs}}) := -C_{\text{max}}.$$

$$\varepsilon_{\text{res}}^* \leq \varepsilon_0 + (1 + \gamma^K) L_V(R) R, \quad (16)$$

with $\varepsilon_0 := \|B_0 - \mathcal{T}_K^\pi B_0\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Projection thus enters both the architectural term (through $L_V(R)R$) and the truncation term (through $\bar{L}_z^{\text{sat}}(R)$ and $C_z \leq 2R$). Any finite-batch measurement is a diagnostic, not a certificate of the uniform ρ_R and Lipschitz assumptions on $\mathcal{C}^\circ$. Likewise, a finite-batch estimate of the self-bootstrap U_n residual is only a proxy: the uniform sup-norm residual enters the upper bound on $\varepsilon_{\text{res}}^*$ together with the truncation gap (Remark G.2).

For the policy-overlap refinements below, define on the current-policy state domain

$$\tau(s) := \text{TV}(\pi_{\text{cand}}(\cdot | s), \pi(\cdot | s)), \quad \tau_\infty := \sup_{s \in \mathcal{C}_\pi} \tau(s) \leq 1. \quad (17)$$

Corollary 6.2 (Self-bootstrap and target-network residual certificates). *Use the measurable Bellman domain of Theorem 6.1 with $\mathcal{C} = \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$. Take $U_n \in B_b(\mathcal{C})$ and use the common scalar boundary $U_n(s_{\text{abs}}) = V^\pi(s_{\text{abs}}) = b$. Define*

$$\varepsilon_{\text{res}, n} \triangleq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (18)$$

Then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res}, n}}{1 - \gamma^K}. \quad (19)$$

If $\hat{A} = \hat{A}_{U_n}$ is the exact one-step centered estimator of Eq. (140), with its transition expectation and baseline sum under π evaluated exactly, then Assumption 6.5 holds with

$$\varepsilon_A \triangleq \frac{2\gamma \varepsilon_{\text{res}, n}}{1 - \gamma^K}. \quad (20)$$

Consequently, under the hypotheses of Theorem 6.7,

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} \frac{\varepsilon_{\text{res}, n}}{1 - \gamma^K} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (21)$$

For a finite depth $q \geq 0$, instead suppose Proposition 5.2 holds on this same domain with $\mathcal{E} = \mathcal{C}$. Then

$$\|U_q - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{R_{\text{targ}, q}^P}{1 - \gamma^K}. \quad (22)$$

For finite depths $0 \leq n < m$, if $U_n, U_m \in B_b(\mathcal{C})$ use the same scalar boundary and the depth- m target-network quantities are established, then

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{R_{\text{targ}, m}^P}{1 - \gamma^K}. \quad (23)$$

For the depth- q target statement, take $\hat{A} = \hat{A}_{U_q}$ from Eq. (140), with its transition expectation and current-policy baseline evaluated exactly, and define $\varepsilon_{A, \text{cand}}$ by Eq. (44) for this estimator. Corollary 6.4 then gives

$$\varepsilon_{A, \text{cand}} \leq 2\gamma\tau_\infty \frac{R_{\text{targ}, q}^P}{1 - \gamma^K}. \quad (24)$$

Consequently, under the exact-mixture and scalar-span hypotheses of Theorem 6.7,

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{(1 - \gamma)(1 - \gamma^K)} R_{\text{targ}, q}^P - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}, \quad (25)$$

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} \left(\|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{R_{\text{targ}, m}^P}{1 - \gamma^K} \right) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (26)$$

The second display uses the same exact value-derived construction at depth n . In all these formulas, $R_{\text{targ}, q}^P$ retains the sharper propagated-lag norm from Eq. (2); $R_{\text{targ}, q}^P \leq R_{\text{targ}, q}^{\text{sup}}$ gives the ordinary sup-lag fallback.

This certificate follows from Bellman contraction alone and applies to measurable state spaces, including continuous augmented latents, under the displayed $B_b(\mathcal{C})$, invariance, measurability, and common-boundary assumptions. Boundedness is essential to this sup-norm statement. For example, on the countable zero-reward chain $s_i \mapsto s_{i+1}$ with one action and $\gamma \in (0, 1)$, plus an unreachable zero-reward absorber with $U(s_{\text{abs}}) = V^\pi(s_{\text{abs}}) = 0$, the unbounded function $U(s_i) = \gamma^{-i}$ has zero K -step residual for every $K \geq 1$ while its sup-norm error from $V^\pi = 0$ is infinite.

The CPI guarantees below are deterministic conditional statements that need sup-norm residual control on $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$. Under its displayed bounded-domain conditions, Corollary 6.2 can certify it when the closure and backup are exactly evaluable; otherwise it remains an assumption. Because the closure depends on both π and π_{cand} , residual control is a property of the policy pair. The certificate is post-candidate: once π_{cand} is chosen, the relevant closure can be checked; future candidates require their own check.

Corollary 6.3 (Advantage-error bridge). *Under the hypotheses of Theorem 6.1, define*

$$B_{n,m} := \|U_n - U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \frac{\|U_m - \mathcal{T}_K^\pi U_m\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}}{1 - \gamma^K}. \quad (27)$$

Let $\hat{A} = \hat{A}_{U_n}$ be the exact one-step centered estimator of Eq. (140), with the transition expectation evaluated exactly (or applied to a deterministic transition) and the baseline computed exactly under π . Then Assumption 6.5 holds with the valid choice

$$\varepsilon_A = 2\gamma B_{n,m}. \quad (28)$$

If Assumption 5.1 holds on $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the L_V -Lipschitz head condition also holds there, sending $m \rightarrow \infty$ gives the fixed-point specialization

$$\varepsilon_A \leq \underbrace{\frac{2\gamma \varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{architectural}} + \underbrace{2\gamma L_V \frac{L_z^n}{1 - L_z}}_{\text{unrolling}} C_z. \quad (29)$$

Corollary 6.4 (Candidate evaluation bias from policy overlap). *Let $\hat{A}_V$ be the exactly centered one-step estimator, $e = V - V^\pi$, and $\delta_V = \|e\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Use τ and τ_∞ from Eq. (17). Then, for every $s \in \mathcal{C}_\pi$,*

$$\left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)} [\hat{A}_V(s, a) - A^\pi(s, a)] \right| \leq 2\gamma \delta_V \tau(s), \quad (30)$$

and hence

$$\varepsilon_{A, \text{cand}} \leq 2\gamma \delta_V \tau_\infty, \quad (31)$$

$$\mathbb{E}_{s \sim d_\pi} |b(s)| \leq 2\gamma \delta_V \mathbb{E}_{s \sim d_\pi} \tau(s). \quad (32)$$

For an exactly centered K -step estimator whose intervening actions follow π , replace γ by γ^K . The fallback $2\gamma \delta_V$ follows from $\tau \leq 1$.

Proof. Let $q_e(s, a) = \gamma \mathbb{E}[e(S') | s, a]$. Exact centering gives $b(s) = \mathbb{E}_{\pi_{\text{cand}}} q_e - \mathbb{E}_\pi q_e$. By Definition D.2, there is a common action set of full $\pi(\cdot | s)$ - and $\pi_{\text{cand}}(\cdot | s)$ -measure on which $q_e(s, a) \in [-\gamma \delta_V, \gamma \delta_V]$. The total-variation expectation inequality on this common essential range proves Eq. (30); taking a supremum or a d_π expectation gives the two corollaries. For a K -step estimator, $q_e^{(K)}(s, a) = \gamma^K \mathbb{E}[e(S_K) | s, a]$ has the corresponding essential range $[-\gamma^K \delta_V, \gamma^K \delta_V]$. $\square$

Here the value-error norm is taken over $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ (Definition D.2). For a centered K -step estimator, 2γ becomes $2\gamma^K$; GAE-style constants can likewise be absorbed into ε_A (Remark H.3). Combining the finite-reference choice with Corollary 6.4 and Theorem 6.7 gives a penalty that does not require a recurrent latent fixed point:

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} B_{n,m} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (33)$$

Thus a finite recurrent path and a residual at any deeper finite depth feed directly into CPI. Geometric L_z^n dependence is only the contractive specialization.

6.2 CONSERVATIVE POLICY IMPROVEMENT

For the exact pointwise probability-space mixture $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$, the Bernoulli mixture selector yields a sharper occupancy constant than a generic policy perturbation argument. On a common domain with a fixed advantage span, the resulting policy-shift term is $O(\alpha^2)$ near $\alpha = 0$, with an exact denominator that depends on α .

Assumption 6.5 (Uniform advantage error on $\mathcal{C}_\pi$). For some $\varepsilon_A \geq 0$:

$$|\hat{A}(s, a) - A^\pi(s, a)| \leq \varepsilon_A \quad (34)$$

for every $s \in \mathcal{C}_\pi$ and for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a .

Write $d_\pi = (1 - \gamma) \sum_{t \geq 0} \gamma^t \rho P_\pi^t$ and define the true and estimated CPI surrogates

$$\begin{aligned} L_\pi(\pi') &:= \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot | s)} [A^\pi(s, a)], \\ \hat{L}_\pi(\pi') &:= \eta(\pi) + \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot | s)} [\hat{A}(s, a)]. \end{aligned}$$

Lemma 6.6 (Exact-mixture occupancy). *Let π and π_{cand} act in the same MDP from the same initial law, and let π_α be their exact pointwise probability mixture. For $\alpha \in [0, 1]$,*

$$\text{TV}(d_{\pi_\alpha}, d_\pi) \leq \frac{\gamma\alpha}{1 - \gamma + \gamma\alpha}. \quad (35)$$

For finite or countable state spaces, with $\text{TV}(p, q) = \frac{1}{2} \|p - q\|_1$,

$$\|d_{\pi_\alpha} - d_\pi\|_1 \leq \frac{2\gamma\alpha}{1 - \gamma + \gamma\alpha}. \quad (36)$$

Proof. Realize each mixture decision with an independent Bernoulli selector. The event that decisions $0, \dots, t - 1$ all select the current-policy component has probability $(1 - \alpha)^t$ and, conditional on that event, produces the current-policy state law before decision t . Therefore

$$\text{TV}(\rho P_{\pi_\alpha}^t, \rho P_\pi^t) \leq 1 - (1 - \alpha)^t.$$

Convexity of total variation and the geometric identity give

$$\begin{aligned} \text{TV}(d_{\pi_\alpha}, d_\pi) &\leq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t [1 - (1 - \alpha)^t] \\ &= \frac{\gamma\alpha}{1 - \gamma + \gamma\alpha}. \end{aligned}$$

The exponent is t , not $t + 1$, because occupancy is measured before decision t . The formula gives zero at $\alpha = 0$ or $\gamma = 0$ and γ at $\alpha = 1$, and improves the generic bound because $1 - \gamma + \gamma\alpha \geq 1 - \gamma$. In the countable case this gives $2\gamma\alpha/(1 - \gamma + \gamma\alpha) \leq 2\gamma\alpha/(1 - \gamma)$. $\square$

Theorem 6.7 (Occupancy-aware CPI with signed evaluation error). *Fix one discounted MDP with bounded rewards and measurable current and candidate policy kernels. Let $\alpha \in [0, 1]$ and let $\pi_\alpha = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$ be their exact pointwise probability-space mixture, with no parameter interpolation, hidden-state interpolation, logit interpolation, or distillation. Let $\mathcal{C}_{\text{mix}}$ be measurable with $\rho(\mathcal{C}_{\text{mix}}) = 1$ and invariant under both π and π_α , as in Definition D.3. Assume that a measurable π -invariant domain $\mathcal{C}_\pi \subseteq \mathcal{C}_{\text{mix}}$ also carries ρ . Assume the functions below are bounded and measurable on their declared domains. Define g on $\mathcal{C}_{\text{mix}}$ and c, b on $\mathcal{C}_\pi$ by*

$$g(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)} [A^\pi(s, a)], \quad \Delta_g := \sup_{s \in \mathcal{C}_{\text{mix}}} g(s) - \inf_{s \in \mathcal{C}_{\text{mix}}} g(s), \quad (37)$$

$$c(s) := \mathbb{E}_{a \sim \pi(\cdot | s)} [\hat{A}(s, a)], \quad b(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot | s)} [\hat{A}(s, a) - A^\pi(s, a)], \quad (38)$$

$$\Xi_\alpha := \mathbb{E}_{s \sim d_\pi} [(1 - \alpha)c(s) + \alpha b(s)]. \quad (39)$$

Then the surrogate error is exactly

$$\hat{L}_\pi(\pi_\alpha) - L_\pi(\pi_\alpha) = \frac{\Xi_\alpha}{1 - \gamma}, \quad (40)$$

and

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{\Xi_\alpha}{1-\gamma} - \frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (41)$$

Because Ξ_α is signed, a usable one-sided guarantee may replace it by any established $B_\alpha \geq \Xi_\alpha$, for example

$$B_\alpha = (1-\alpha)\mathbb{E}_{d_\pi}|c(s)| + \alpha\mathbb{E}_{d_\pi}|b(s)|. \quad (42)$$

Under exact statewise centering,

$$\mathbb{E}_{a \sim \pi(\cdot|s)}[\hat{A}(s, a)] = 0 \quad \text{for every } s \in \mathcal{C}_\pi, \quad (43)$$

define the uniform candidate-policy bias

$$\varepsilon_{A, \text{cand}} \triangleq \sup_{s \in \mathcal{C}_\pi} \left| \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}(s, a) - A^\pi(s, a)] \right|. \quad (44)$$

If $\varepsilon_{A, \text{cand}} < \infty$ and $\varepsilon_{\text{CPI}} := \sup_{s \in \mathcal{C}_{\text{mix}}} |g(s)| < \infty$, then

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{\alpha}{1-\gamma} \varepsilon_{A, \text{cand}} - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}, \quad (45)$$

Theorem 6.7 is pointwise in α : its $\mathcal{C}_{\text{mix}}$ and the resulting span Δ_g need only be valid for the selected mixture weight and may depend on that choice. An interval certificate instead fixes the common policy-pair domain and all constants as in Corollary 6.8.

Assumption 6.5 is one sufficient condition implying $\varepsilon_{A, \text{cand}} \leq \varepsilon_A$; it is not required for the signed master inequality. Equation (43) holds when the estimator is constructed by subtracting the exact policy expectation. In a finite action space this can be an explicit sum; on a general action space it is an exact kernel integral. Sampling-based or learned baselines instead enter through $c(s)$ or through an upper bound B_α .

The constants split into estimable and assumed quantities. Local finite-difference and batch measurements are diagnostics, not uniform bounds. In particular, the architectural residual $\varepsilon_{\text{res}}^*$ is a sup norm over the advantage closure; a uniform self-bootstrap residual relates to it through the truncation gap in Remark G.2, while a target-network backup first uses Proposition 5.2. A finite diagnostic population cannot by itself calibrate any of these uniform quantities.

Corollary 6.8 (Span-based safe mixture step). *In the setting of Theorem 6.7, fix one measurable common policy-pair domain $\mathcal{C}_{\text{pair}}$ from Definition D.3, with $\mathcal{C}_\pi \subseteq \mathcal{C}_{\text{pair}}$. Use one fixed estimator $\hat{A}$ satisfying exact statewise centering on $\mathcal{C}_\pi$ for every $\alpha \in [0, 1]$, and define its fixed candidate-defect function*

$$b(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}(s, a) - A^\pi(s, a)]. \quad (46)$$

Neither the estimator nor this defect object depends on α . Define, once on $\mathcal{C}_{\text{pair}}$,

$$g(s) := \mathbb{E}_{a \sim \pi_{\text{cand}}(\cdot|s)} [A^\pi(s, a)], \quad \Delta_g := \sup_{s \in \mathcal{C}_{\text{pair}}} g(s) - \inf_{s \in \mathcal{C}_{\text{pair}}} g(s), \quad (47)$$

and assume this function is bounded and measurable there. Let $\bar{\beta}_d$ be any finite, α -independent, established one-sided bound satisfying

$$\bar{\beta}_d \geq \mathbb{E}_{s \sim d_\pi} b(s). \quad (48)$$

Define

$$\hat{g} := \mathbb{E}_{s \sim d_\pi, a \sim \pi_{\text{cand}}(\cdot|s)} [\hat{A}(s, a)], \quad M_d := \hat{g} - \bar{\beta}_d. \quad (49)$$

Then, simultaneously for every $\alpha \in [0, 1]$,

$$\eta(\pi_\alpha) - \eta(\pi) \geq \frac{\alpha}{1-\gamma} \left[M_d - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right]. \quad (50)$$

Thus $\eta(\pi_\alpha) \geq \eta(\pi)$ is certified whenever

$$M_d \geq \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha}. \quad (51)$$

Three immediate choices preserve the same fixed defect object:

1. the exact signed defect, $\bar{\beta}_d = \mathbb{E}_{d_\pi} b(s)$;
2. the occupancy-averaged absolute defect, $\bar{\beta}_d = \mathbb{E}_{d_\pi} |b(s)|$;
3. the uniform specialization already used above, $\bar{\beta}_d = \varepsilon_{A,\text{cand}}$ from Eq. (44).

The third choice recovers the previous uniform statement as an immediate specialization. The zero step $\alpha = 0$ is always equality. At $\alpha = 1$, this bound certifies the full candidate step exactly when $M_d \geq \gamma \Delta_g$. The remaining sharp nonredundant boundary cases for positive steps are:

- if $\gamma = 0$ and $M_d \geq 0$, every $\alpha \in [0, 1]$ is certified;
- if $M_d < 0$, no positive step is certified by this bound;
- if $\gamma > 0$, $M_d = 0$, and $\Delta_g > 0$, no positive step is certified;
- if $M_d = 0$ and $\Delta_g = 0$, every step is certified with a zero lower bound;
- if $\gamma > 0$, $M_d > 0$, and $M_d \geq \gamma \Delta_g$, every $\alpha \in [0, 1]$ is certified; the equality case $M_d = \gamma \Delta_g$ belongs to this all-step branch;
- if $\gamma > 0$ and $0 < M_d < \gamma \Delta_g$, every

$$\alpha \leq \frac{M_d(1 - \gamma)}{\gamma(\Delta_g - M_d)} \quad (52)$$

is certified. Here $\Delta_g - M_d > 0$, and the displayed threshold lies strictly between zero and one.

Under Corollaries 6.3 and 6.4, with the finite-reference estimator $\hat{A} = \hat{A}_{U_n}$, one may use $\mathbb{E}_{d_\pi} |b(s)| \leq 2\gamma B_{n,m} \mathbb{E}_{d_\pi} \tau(s)$ for the occupancy-averaged specialization, or $\varepsilon_{A,\text{cand}} \leq 2\gamma B_{n,m} \tau_\infty$ for the uniform specialization. The bound $\tau_\infty \leq 1$ recovers the previous uniform fallback.

Theorem 6.7 applies to the exact mixture; distillation and parameter interpolation are different policies. They are covered only after a deployment discrepancy is established. The policy-shift term is $O(\alpha^2)$ near zero when its domain and span are held fixed, not a polynomial quadratic for all α .

Corollary 6.9 (One-step parameter dependence). *Under Corollaries 6.3 and 6.4 and Theorem 6.7,*

$$\eta(\pi_{\text{new}}) \geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} \left(\frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K} + L_V \frac{L_z^n}{1 - L_z} C_z \right) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \quad (53)$$

Corollary 6.9 is the optional fixed-point specialization of the overlap-aware finite-reference penalty $2\alpha\gamma\tau_\infty B_{n,m}/(1 - \gamma)$. It makes the one-step dependence on (L_z, n, K, α) explicit. For $n \geq 1$, its recurrent term vanishes as $L_z \rightarrow 0$; together with $\varepsilon_{\text{res}}^* \rightarrow 0$, this recovers the exact-mixture CPI expression. At $n = 0$, the displayed recurrent term instead tends to $L_V C_z$ and no such claim is made. Finite-difference values are local monitoring quantities and certify no uniform modulus on the declared domain.

A projection-aware version of (53) substitutes $\varepsilon_0 + (1 + \gamma^K) L_V(R) R$ for $\varepsilon_{\text{res}}^*$ and the conditional analytic upper bound $\bar{L}_z^{\text{sat}}(R)$ for L_z ; the full statement is Eq. (143) in Appendix I.5. The saturated-regime upper bound depends multiplicatively on both R/ρ_R and $L_z^{\text{pre}}(R)$, and local proxies certify neither uniform domain-relative factor. For $n \geq 1$, a smaller established uniform upper bound on L_z or larger n reduces L_z^n ; holding a common domain and the displayed bounds fixed, smaller α is safer; larger K tightens $1/(1 - \gamma^K)$ at the cost of a K -dependent residual. Persistent- z uses the direct augmented-state residual of Appendix B; slow drift enters only the optional comparison to a memoryless fixed-point reference.

7 DISCUSSION, LIMITATIONS, AND CONCLUSION

The main result is a fixed-snapshot decomposition on $B_b(\mathcal{C})$: a finite-depth recurrent evaluator is compared with any deeper finite reference through a recurrent-path term and a reference-depth Bellman residual. The finite-horizon version uses the remaining edit clock and stagewise advantage spans. Contraction is optional. Persistent latents use the clock-complete state (x, y, z, h) and do not

require a recurrent latent fixed point, while the Bellman argument still uses the value-function fixed point.

Exact probability mixing yields the occupancy denominator $1 - \gamma + \gamma\alpha$ and exposes signed, occupancy-averaged centering and candidate defects. Policy overlap reduces value-derived candidate bias. A different deployed policy incurs the first-mismatch denominator $1 - \gamma + \gamma\delta_{\text{dep}}$ when its uniform discrepancy from the exact mixture is established on a recursively invariant deployment domain.

The paper includes no empirical evaluation. The results do not analyze optimization dynamics and do not establish that UPI-TRM improves task performance. Any later learned-model claim must separately establish the uniform theorem premises or be labeled as a finite-batch diagnostic. In particular, a fixed-target population residual reaches the value and CPI certificates only after adding its propagated target lag; it is not the self-bootstrap Bellman residual. The ordinary target sup-lag is a looser fallback. The same fixed recurrent map is required when current and candidate policies share one augmented MDP. A small Euclidean difference between two recurrent maps is insufficient for a return perturbation bound; a different-MDP result needs explicit kernel/reward discrepancies or a separate Wasserstein analysis.

REFERENCES

- Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, volume 32, pp. 690–701, 2019.
- Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming*. Athena Scientific, 1996. ISBN 1886529108.
- Kai-Wei Chang, Akshay Krishnamurthy, Alekh Agarwal, Hal Daumé III, and John Langford. Learning to search better than your teacher. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 2058–2066. PMLR, 2015.
- Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. In *Advances in Neural Information Processing Systems*, volume 34, pp. 15084–15097, 2021.
- Hal Daumé III, John Langford, and Daniel Marcu. Search-based structured prediction. *Machine Learning*, 75(3):297–325, 2009.
- DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning, 2025. URL <https://arxiv.org/abs/2501.12948>.
- S. R. Eshwar, Gagan Thoppe, Ananyabrata Barua, Aditya Gopalan, and Gal Dalal. Monotone and conservative policy iteration beyond the tabular case, 2025. URL <https://arxiv.org/abs/2506.07134>.
- Michael Janner, Qiyang Li, and Sergey Levine. Offline reinforcement learning as one big sequence modeling problem. In *Advances in Neural Information Processing Systems*, volume 34, pp. 1273–1286, 2021.
- Alexia Jolicoeur-Martineau. Less is more: Recursive reasoning with Tiny networks. *arXiv preprint arXiv:2510.04871*, 2025.
- Sham Kakade and John Langford. Approximately optimal approximate reinforcement learning. In *Proceedings of the 19th International Conference on Machine Learning*, pp. 267–274, 2002.
- Kevin Klein. Sudoku #3: Poor man’s RL. Blog post, 2022. Accessed 2026-01-09.
- Michael Laskin, Luyu Wang, Junhyuk Oh, Emilio Parisotto, Stephen Spencer, Richie Steigerwald, DJ Strouse, Steven Stenberg Hansen, Angelos Filos, Ethan Brooks, Maxime Gazeau, Himanshu Sahni, Satinder Singh, and Volodymyr Mnih. In-context reinforcement learning with algorithm distillation. In *International Conference on Learning Representations*, 2023.
- Ronan Killian McGovern. Test-time adaptation of tiny recursive models. *arXiv preprint arXiv:2511.02886*, 2025.

-
- Anav Mehta. Reinforcement learning for constraint satisfaction game agents (15-puzzle, minesweeper, 2048, and sudoku). *arXiv preprint arXiv:2102.06019*, 2021. URL <https://arxiv.org/abs/2102.06019>.
- Alberto Maria Metelli, Matteo Pirota, Daniele Calandriello, and Marcello Restelli. Safe policy iteration: A monotonically improving approximate policy iteration approach. *Journal of Machine Learning Research*, 22(97):1–83, 2021.
- Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the Sixteenth International Conference on Machine Learning*, pp. 278–287. Morgan Kaufmann, 1999.
- Matteo Pirota, Marcello Restelli, Alessio Pecorino, and Daniele Calandriello. Safe policy iteration. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pp. 307–315. PMLR, 2013.
- Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley-Interscience, 1994. ISBN 0471619779.
- Kaleem Ullah Qasim and Jiashu Zhang. Accelerating training speed of tiny recursive models with curriculum guided adaptive recursion. *arXiv preprint arXiv:2511.08653*, 2025.
- Stephane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pp. 627–635. PMLR, 2011.
- John Schulman, Sergey Levine, Philipp Moritz, Michael I. Jordan, and Pieter Abbeel. Trust region policy optimization. In *Proceedings of the 32nd International Conference on Machine Learning*, pp. 1889–1897, 2015.
- Yi Su, Dian Yu, Linfeng Song, Juntao Li, Haitao Mi, Zhaopeng Tu, Min Zhang, and Dong Yu. Crossing the reward bridge: Expanding rl with verifiable rewards across diverse domains, 2025. URL <https://arxiv.org/abs/2503.23829>.
- Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.
- Aviv Tamar, Yi Wu, Garrett Thomas, Sergey Levine, and Pieter Abbeel. Value iteration networks. In *Advances in Neural Information Processing Systems*, volume 29, 2016.
- Guan Wang, Jin Li, Yuhao Sun, Xing Chen, Changling Liu, Yue Wu, Meng Lu, Sen Song, and Yasin Abbasi-Yadkori. Hierarchical Reasoning Model. *arXiv preprint arXiv:2506.21734*, 2025a.
- Yiping Wang, Qing Yang, Zhiyuan Zeng, Liliang Ren, Liyuan Liu, Baolin Peng, Hao Cheng, Xuehai He, Kuan Wang, Jianfeng Gao, Weizhu Chen, Shuohang Wang, Simon Shaolei Du, and Yelong Shen. Reinforcement learning for reasoning in large language models with one training example, 2025b. URL <https://arxiv.org/abs/2504.20571>.

A SUPPLEMENTARY MATERIAL

Assumption A.1 (Bounded rewards). Potential-shaped rewards are uniformly bounded. For some $R_{\max} < \infty$,

$$|r(s, a, s')| \leq R_{\max} \quad \text{for every transition.}$$

For $R > 0$, define the Euclidean projection

$$\begin{aligned} \Pi_R(z) &= \arg \min_{u: \|u\|_2 \leq R} \|u - z\|_2 \\ &= \begin{cases} z, & \|z\|_2 \leq R, \\ R \frac{z}{\|z\|_2}, & \|z\|_2 > R. \end{cases} \end{aligned} \quad (54)$$

$$G^{(K)}(s_0) = \sum_{k=0}^{K-1} \gamma^k r_k + \gamma^K \bar{V}(s_K). \quad (55)$$

$$C_z := \sup_{s \in \mathcal{C}^\circ} \|\tilde{f}_\theta(z_{\text{init}}(x(s), y(s)), y(s), x(s)) - z_{\text{init}}(x(s), y(s))\| < \infty. \quad (56)$$

Proposition A.2 (Domain-relative fixed point and finite-unrolling bias (Banach)). *Under Assumption 5.1, for every $s \in \mathcal{D}^\circ$ the map T_s has a unique fixed point $z^*(s) \in \mathcal{Z}_{\text{inv}}$. For every $z^{(0)}(s) \in \mathcal{Z}_{\text{inv}}$, its iterates converge linearly:*

$$\|z^{(t)}(s) - z^*(s)\| \leq L_z^t \|z^{(0)}(s) - z^*(s)\|. \quad (57)$$

If $\mathcal{D}^\circ = \mathcal{C}^\circ$, C_z is finite as in Eq. (56), and $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$, then uniformly on $\mathcal{C}^\circ$,

$$\|z^{(n)}(s) - z^*(s)\| \leq \frac{L_z^n}{1 - L_z} C_z. \quad (58)$$

Thus fixed-point existence is asserted only for input pairs represented by the declared domain. Assumption 5.1's metric contraction and invariance clauses alone do not imply measurability of $s \mapsto z^*(s)$.

For a sufficient measurability condition, equip the closed set $\mathcal{Z}_{\text{inv}}$ with its Borel σ -algebra and $\mathcal{D}^\circ$ with its declared σ -algebra. If

$$T : \mathcal{D}^\circ \times \mathcal{Z}_{\text{inv}} \longrightarrow \mathcal{Z}_{\text{inv}}, \quad T(s, z) = T_s(z), \quad (59)$$

is jointly measurable and $z^{(0)} : \mathcal{D}^\circ \rightarrow \mathcal{Z}_{\text{inv}}$ is measurable, then every iterate $z^{(j)}$ and the fixed-point map z^* are measurable.

For any result below that invokes L_V on a declared nonabsorbing domain $\mathcal{D}^\circ$, let $\mathcal{Z}_{\mathcal{D}}(s)$ be the set of latent arguments compared by that result at s . Assume the scalar head is uniformly Lipschitz while the instance and plan are held fixed: for every $s \in \mathcal{D}^\circ$ and all $z, z' \in \mathcal{Z}_{\mathcal{D}}(s)$,

$$|V_\psi(z, x(s), y(s)) - V_\psi(z', x(s), y(s))| \leq L_V \|z - z'\|. \quad (60)$$

For a contractive specialization one may take $\mathcal{Z}_{\mathcal{D}}(s) = \mathcal{Z}_{\text{inv}}$; a pathwise result needs the condition only at the latent pairs it displays.

Assumption A.3 (Domain-explicit bounded value heads). For an invocation on a declared nonabsorbing state domain $\mathcal{D}^\circ$, declare a latent set $\mathcal{Z}_{\text{bd}} \subseteq \mathcal{Z}_{\text{inv}}$. It is either all of $\mathcal{Z}_{\text{inv}}$ or an explicitly identified smaller set containing every latent used by that result. There exists $V_{\max} < \infty$ such that, for every represented $s \in \mathcal{D}^\circ$ and every $z \in \mathcal{Z}_{\text{bd}}$,

$$|V_\psi(z, x(s), y(s))| \leq V_{\max}, \quad |V_{\bar{\psi}}(z, x(s), y(s))| \leq V_{\max}. \quad (61)$$

The absorber is handled by its separately declared finite scalar boundary; no absorbing latent is introduced. This assumption can supply boundedness in the optional contractive specializations, but it does not replace any explicit $B_b(\mathcal{C})$ premise in a Bellman theorem.

For $\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$:

$$\begin{aligned}\eta(\pi_{\text{new}}) &\geq L_\pi(\pi_{\text{new}}) - \frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)} \\ &\geq L_\pi(\pi_{\text{new}}) - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)},\end{aligned}\tag{62}$$

where g , Δ_g , and ε_{CPI} use the $\mathcal{C}_{\text{mix}}$ -restricted definitions in Theorem 6.7.

$$\pi_{\text{new}} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}, \quad \alpha \in [0, 1].\tag{63}$$

B PERSISTENT- z ON THE AUGMENTED MDP

B.1 DIRECT FINITE-DEPTH RESIDUAL CERTIFICATE

The formal statement below applies Theorem 6.1 on a measurable augmented domain. In particular, it permits a continuous carried latent $z \in \mathbb{R}^d$ without reducing it to a countable reachable set.

Fix n , (θ, ψ) , the policy parameters, and all environment kernels. Let $h \in \{1, 2, \dots, H\}$ be the number of edits remaining before the current action. Define

$$F_n(x, y, z) := T_{x,y}^{\circ n}(z) = \tilde{f}_\theta^{\circ n}(z, y, x), \quad \bar{s} = (x, y, z, h),\tag{64}$$

For a nonabsorbing state, first compute F_n and sample an action. If the environment transition terminates for any reason (including a solution or terminal STOP), or if $h = 1$, the successor is the shared atom s_{abs} ; budget exhaustion is terminal. Otherwise the clock-complete successor is

$$\bar{s}' = (x, \text{edit}(y, a; x), F_n(x, y, z), h - 1).$$

Thus the state stores the pre-unroll carried latent z , the policy uses the post-unroll latent $F_n(x, y, z)$, and a nonterminal successor carries that post-unroll latent. The current and candidate action kernels share this same frozen F_n and hence the same augmented MDP:

$$\bar{\pi}(a \mid \bar{s}) = \pi_\phi(a \mid y, F_n(x, y, z), x), \quad \bar{\pi}_{\text{cand}}(a \mid \bar{s}) = \pi_{\text{cand}}(a \mid y, F_n(x, y, z), x).$$

Changing the recurrent map between these two policies would change the transition kernel and is outside ordinary CPI.

Remark B.1 (Why the recurrent map is shared). A small Euclidean discrepancy between deterministic recurrent maps does not imply a small policy or augmented-kernel discrepancy. For example, take $F_\varepsilon(z) = -\varepsilon/2$ and $G_\varepsilon(z) = \varepsilon/2$ and a fixed threshold policy head that chooses disjoint actions below and above zero. Then $\sup_z \|F_\varepsilon(z) - G_\varepsilon(z)\| = \varepsilon$, but the action laws are disjoint and a fixed MDP can have an order- $1/(1-\gamma)$ return gap. Moreover, $\text{TV}(\delta_{F(z)}, \delta_{G(z)}) = 1$ whenever $F(z) \neq G(z)$. A different-map theorem therefore needs explicit reward and transition-kernel discrepancies, or a Wasserstein analysis with policy and reward Lipschitz regularity. Map distance alone is insufficient.

The initial law is

$$\bar{\rho}(dx, dy, dz, dh) = \rho(dx, dy, dh) \delta_{z_{\text{init}}(x,y)}(dz),\tag{65}$$

or, for stochastic initialization, $\bar{\rho}(dx, dy, dz, dh) = \rho(dx, dy, dh)\nu_0(dz \mid x, y, h)$ with any recorded measurable initialization kernel. Ignoring h is an architectural specialization, not a Bellman requirement.

Assume the initialization, edit/reward kernels, F_n , V_ψ , and both policy kernels are measurable, and the K -step expected reward and evaluator are bounded. Let $\bar{\mathcal{C}}_\pi$ be a measurable $\bar{\pi}$ -invariant domain with $\bar{\rho}(\bar{\mathcal{C}}_\pi) = 1$. Let $\bar{\mathcal{C}}$ be a measurable advantage-evaluation domain containing $\bar{\mathcal{C}}_\pi$ and the shared absorber, containing the one-step successors from $\bar{\mathcal{C}}_\pi$ for $\bar{\pi}$ - and $\bar{\pi}_{\text{cand}}$ -almost every action, and invariant under every subsequent $\bar{\pi}$ step. In particular, $\bar{\rho}(\bar{\mathcal{C}}) = 1$. Write $\bar{\mathcal{C}}^\circ := \bar{\mathcal{C}} \setminus \{s_{\text{abs}}\}$ for its nonabsorbing part. Keep this augmented MDP, including its deployed depth n , fixed. For any finite comparison depth $q \geq 0$, define the endpoint function

$$\tilde{U}_q(\bar{s}) := V_\psi(\tilde{f}_\theta^{\circ q}(z, y, x), x, y), \quad \tilde{U}_q(s_{\text{abs}}) := -C_{\text{max}}.\tag{66}$$

Changing q here changes only a comparison function on the already-fixed state space. It does not change F_n , either policy kernel, or the augmented transition kernel. Without this restriction, two endpoint depths would belong to different MDPs and no finite-reference comparison would follow. At $q = n$, Eq. (66) is the deployed evaluator. Let $\bar{T}_K^\pi$ denote the K -step Bellman operator of this frozen augmented MDP. Its direct residual is

$$\bar{\varepsilon}_{\text{res},n} := \|\tilde{U}_n - \bar{T}_K^\pi \tilde{U}_n\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (67)$$

Here $\eta(\bar{\pi}')$ denotes the discounted return of any augmented policy $\bar{\pi}'$ from $\bar{\rho}$, and $d_{\bar{\pi}}$ is the discounted occupancy of the current policy. For an augmented advantage estimator $\hat{A}$, define

$$\hat{L}_{\bar{\pi}}(\bar{\pi}') := \eta(\bar{\pi}) + \frac{1}{1-\gamma} \mathbb{E}_{\bar{s} \sim d_{\bar{\pi}}, a \sim \bar{\pi}'(\cdot|\bar{s})} [\hat{A}(\bar{s}, a)]. \quad (68)$$

Corollary B.2 (Direct persistent- z certificate on the augmented MDP). *Let $\bar{\mathcal{C}}$ be the preceding measurable augmented domain and equip $B_b(\bar{\mathcal{C}})$ with its ordinary pointwise sup norm. Under the preceding frozen-MDP, measurability, bounded- K -step-reward, common-boundary, and invariance assumptions, take $\tilde{U}_n \in B_b(\bar{\mathcal{C}})$. Theorem 6.1 gives a unique $V^\pi \in B_b(\bar{\mathcal{C}})$ and*

$$\|\tilde{U}_n - V^\pi\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \frac{\bar{\varepsilon}_{\text{res},n}}{1-\gamma^K}. \quad (69)$$

If $\hat{A}_{\tilde{U}_n}$ uses the exact transition expectation and exact baseline and is therefore centered pointwise under $\bar{\pi}$ on the full augmented state, then Assumption 6.5 holds for every $\bar{s} \in \bar{\mathcal{C}}_\pi$ and for $\bar{\pi}$ - and $\bar{\pi}_{\text{cand}}$ -almost every action, with the valid choice

$$\bar{\varepsilon}_A := \frac{2\gamma \bar{\varepsilon}_{\text{res},n}}{1-\gamma^K}. \quad (70)$$

If, additionally, the analyzed policy is the pointwise exact mixture $\bar{\pi}_{\text{new}} = (1-\alpha)\bar{\pi} + \alpha\bar{\pi}_{\text{cand}}$ with no parameter interpolation or distillation, let $\bar{\mathcal{C}}_{\text{mix}}$ be a measurable augmented domain carrying $\bar{\rho}$, containing $\bar{\mathcal{C}}_\pi$, and invariant under both $\bar{\pi}$ and $\bar{\pi}_{\text{new}}$. Define

$$\bar{\varepsilon}_{\text{CPI}} := \sup_{\bar{s} \in \bar{\mathcal{C}}_{\text{mix}}} |\mathbb{E}_{a \sim \bar{\pi}_{\text{cand}}(\cdot|\bar{s})} A^{\bar{\pi}}(\bar{s}, a)|, \quad (71)$$

$$\bar{\tau}_\infty := \sup_{\bar{s} \in \bar{\mathcal{C}}_\pi} \text{TV}(\bar{\pi}_{\text{cand}}(\cdot|\bar{s}), \bar{\pi}(\cdot|\bar{s})), \quad (72)$$

where $\bar{\mathcal{C}}_\pi \subseteq \bar{\mathcal{C}} \cap \bar{\mathcal{C}}_{\text{mix}}$ is the common full-measure current-policy domain. Then Theorem 6.7 gives

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_\infty}{1-\gamma} \frac{\bar{\varepsilon}_{\text{res},n}}{1-\gamma^K} - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (73)$$

For the target-network analogue, let $\bar{r}_K^\pi$ be the K -step expected reward of this same fixed augmented MDP. Suppose the conditional target trajectory follows $\bar{\pi}$, and take $\tilde{U}_q, \bar{V} \in B_b(\bar{\mathcal{C}})$ with the common scalar absorbing boundary. Define

$$\bar{\varepsilon}_{\text{targ},q} := \left\| \tilde{U}_q - (\bar{r}_K^\pi + \gamma^K P_{\bar{\pi}}^K \bar{V}) \right\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}, \quad (74)$$

$$\bar{R}_{\text{targ},q}^P := \bar{\varepsilon}_{\text{targ},q} + \gamma^K \|P_{\bar{\pi}}^K (\bar{V} - \tilde{U}_q)\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}, \quad (75)$$

$$\bar{R}_{\text{targ},q}^{\text{sup}} := \bar{\varepsilon}_{\text{targ},q} + \gamma^K \|\bar{V} - \tilde{U}_q\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (76)$$

Then

$$\|\tilde{U}_q - \bar{T}_K^\pi \tilde{U}_q\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \bar{R}_{\text{targ},q}^P \leq \bar{R}_{\text{targ},q}^{\text{sup}}, \quad (77)$$

$$\|\tilde{U}_q - V^\pi\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \frac{\bar{R}_{\text{targ},q}^P}{1-\gamma^K}. \quad (78)$$

For fixed-MDP endpoint depths $0 \leq n < m$,

$$\|\tilde{U}_n - V^\pi\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq \|\tilde{U}_n - \tilde{U}_m\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} + \frac{\bar{R}_{\text{targ},m}^P}{1-\gamma^K}. \quad (79)$$

Let $\hat{A}_{\tilde{U}_q}$ be the one-step estimator of Eq. (140) on the augmented MDP, with its transition expectation and current-policy baseline evaluated exactly conditional on the full augmented state. Define

$$\bar{\varepsilon}_{A,\text{cand}} := \sup_{\bar{s} \in \bar{\mathcal{C}}_\pi} \left| \mathbb{E}_{a \sim \bar{\pi}_{\text{cand}}(\cdot|\bar{s})} \left[\hat{A}_{\tilde{U}_q}(\bar{s}, a) - A^{\bar{\pi}}(\bar{s}, a) \right] \right|. \quad (80)$$

Exact baseline evaluation centers this estimator conditional on the full augmented state, and the candidate bias satisfies

$$\bar{\varepsilon}_{A,\text{cand}} \leq 2\gamma\bar{\tau}_\infty \frac{\bar{R}_{\text{targ},q}^P}{1 - \gamma^K}. \quad (81)$$

The target-network CPI and finite-reference CPI forms are therefore

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_\infty}{(1-\gamma)(1-\gamma^K)} \bar{R}_{\text{targ},q}^P - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}, \quad (82)$$

$$\eta(\bar{\pi}_{\text{new}}) \geq \hat{L}_{\bar{\pi}}(\bar{\pi}_{\text{new}}) - \frac{2\alpha\gamma\bar{\tau}_\infty}{1-\gamma} \left(\|\tilde{U}_n - \tilde{U}_m\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} + \frac{\bar{R}_{\text{targ},m}^P}{1-\gamma^K} \right) - \frac{2\bar{\varepsilon}_{\text{CPI}}\gamma\alpha^2}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (83)$$

The last display uses the same exact value-derived construction at $\tilde{U}_n$. Every target formula keeps the propagated lag before the ordinary sup-lag fallback.

This direct result is Bellman contraction on the actual finite-depth augmented evaluator. It requires no recurrent latent fixed point, recurrent contraction, Lipschitz head, projection, or slow-drift assumption; it still uses the value-function fixed point $V^{\bar{\pi}}$. The augmented residual differs from the episodic residual on (x, y, h) because both the policy and transition can depend on the carried latent. A learned instantiation would additionally need to establish the uniform augmented residual, exact augmented-state centering, and pointwise exact-mixture deployment. Empirical regression loss, a finite-batch maximum, and population L^2 error establish neither the target population sup norm nor either required target-lag sup norm.

Remark B.3 (Scope of the direct augmented-state certificate). Corollary B.2 is a result on $\bar{s} = (x, y, z, h)$, not on the plan marginal (x, y) . Its exact-centering condition averages actions conditional on the full $\bar{s}$, and its mixture identity must hold pointwise on the augmented mixture-occupancy domain $\bar{\mathcal{C}}_{\text{mix}}$. Finite-batch maxima are diagnostics, not uniform certificates.

Lemma B.4 (Finite-path carried-latent comparison). Let $z_{t+1} = F_t(z_t)$ for $t = 0, \dots, T-1$, and let $\xi_0, \dots, \xi_T$ be an arbitrary reference path. Suppose along these paths that, for every $t = 0, \dots, T-1$,

$$\|F_t(z_t) - F_t(\xi_t)\| \leq \kappa_t \|z_t - \xi_t\|, \quad d_t := \|F_t(\xi_t) - \xi_{t+1}\|.$$

Then, with empty products equal to one,

$$\|z_t - \xi_t\| \leq \|z_0 - \xi_0\| \prod_{i=0}^{t-1} \kappa_i + \sum_{j=0}^{t-1} d_j \prod_{i=j+1}^{t-1} \kappa_i. \quad (84)$$

If a head evaluated directly at z_t is $L_{V,t}$ -Lipschitz, its error is at most $L_{V,t}$ times the right-hand side. In the architecture used here, the state stores the pre-unroll z_t and the head reads $F_t(z_t)$; relative to $F_t(\xi_t)$ its error is therefore at most $L_{V,t}\kappa_t\|z_t - \xi_t\|$.

Proof. Writing $e_t = \|z_t - \xi_t\|$, the triangle inequality gives $e_{t+1} \leq \kappa_t e_t + d_t$. Induction gives Eq. (84); head Lipschitzness gives the final claims. This is a pathwise statement. It becomes a uniform sup-norm certificate only when the initial error, κ_t , d_t , and $L_{V,t}$ are bounded uniformly over every declared nonabsorbing start and trajectory. $\square$

B.2 OPTIONAL SLOW-DRIFT COMPARISON TO A FIXED-POINT REFERENCE

Slow drift is needed only for the stronger architectural comparison between the actual transient evaluator and the memoryless reference $\tilde{U}_*(x, y, z, h) := V_\psi(z^*(x, y), x, y)$, with $\tilde{U}_*(s_{\text{abs}}) = -C_{\text{max}}$. For this subsection, invoke Assumption 5.1 with the declared domain $\mathcal{D}^\circ = \bar{\mathcal{C}}^\circ$. Thus $z^*(x, y)$ denotes the unique fixed point only for input pairs represented by this augmented closure;

no claim is made about other pairs. Assume $(\bar{s}, u) \mapsto \tilde{f}_\theta(u, y(\bar{s}), x(\bar{s}))$ is jointly measurable on $\bar{\mathcal{C}}^\circ \times \mathcal{Z}_{\text{inv}}$ and impose a measurable iteration initialization $z^{(0)} : \bar{\mathcal{C}}^\circ \rightarrow \mathcal{Z}_{\text{inv}}$. Proposition A.2 then makes the fixed-point map measurable. Invoke Assumption A.3 with $\mathcal{D}^\circ = \bar{\mathcal{C}}^\circ$ and $\mathcal{Z}_{\text{bd}} = \mathcal{Z}_{\text{inv}}$; with the separate finite absorbing boundary, $\tilde{U}_*$ is bounded and measurable on the augmented closure. Assume the tracking conditions below hold uniformly over nonabsorbing trajectory segments formed by one action drawn under either policy followed by $\bar{\pi}$ dynamics, and assume every carried latent on those segments lies in $\mathcal{Z}_{\text{inv}}$. These segments include $\bar{\rho}$ -almost every initial augmented state and every starting state used to generate a one-deviation trajectory in the augmented closure. In the stochastic-initialization case, require $\nu_0(\mathcal{Z}_{\text{inv}} \mid x, y, h) = 1$ for ρ -almost every (x, y, h) . Let $e : \mathcal{Y} \rightarrow \mathcal{E}_y$ be a fixed plan embedding and write $\|y - y'\|_\mathcal{E} := \|e(y) - e(y')\|_{\mathcal{E}_y}$.

Assumption B.5 (Slow drift of fixed points). Uniformly over the nonabsorbing augmented trajectory segments declared above, there exist $L_{z^*} \geq 0$, $\Delta y_{\text{max}} \geq 0$, and $E_0 < \infty$ such that

1. for every consecutive source/successor pair with common instance x and represented plans y_t, y_{t+1} , $\|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq L_{z^*} \|y_t - y_{t+1}\|_\mathcal{E}$;
2. every admissible edit satisfies $\|y_{t+1} - y_t\|_\mathcal{E} \leq \Delta y_{\text{max}}$; and
3. for $\bar{\rho}$ -almost every (x, y_0, z_0, h_0) , and uniformly for every augmented state treated as time zero when constructing a one-deviation trajectory in the policy-pair closure, the carried latent satisfies $\|z_0 - z^*(x, y_0)\| \leq E_0$.

Remark B.6 (A conditional sufficient bound for L_{z^*}). If y has a specified differentiable continuous representation, f_θ is C^1 near an interior fixed point, and $\|\partial_z f_\theta\|_{\text{op}} \leq L_z < 1$, the implicit-function calculation gives the local derivative bound

$$\|D_y z^*(x, y)\|_{\text{op}} \leq \frac{\|D_y f_\theta(z^*(x, y), y, x)\|_{\text{op}}}{1 - L_z}. \quad (85)$$

This pointwise statement does not by itself supply the single uniform L_{z^*} required by Assumption B.5. For every represented consecutive plan pair, that conclusion additionally needs a C^1 fixed-point branch along a connecting path and one derivative bound uniform over all such paths. For discrete Sudoku plans, the manuscript provides and verifies no such extension.

Remark B.7 (Discrete-plan scope). For a discrete domain, Assumption B.5 is simply an additional regularity assumption in the stated embedding. Successful training does not by itself establish it, and finite-batch diagnostics do not establish the uniform assumption.

Remark B.8 (Scope of the slow-drift assumption). Only the augmented closure for the frozen policy pair is relevant; no global smoothness claim outside that closure is needed.

For this optional subsection fix $n \geq 1$. At nonabsorbing times define $e_t = \|z_t - z^*(x, y_t)\|$ and $\kappa_n = L_z^n < 1$.

Lemma B.9 (Drift bound for persistent- z). *Under the $\mathcal{D}^\circ = \bar{\mathcal{C}}^\circ$ invocation of Assumption 5.1 and Assumption B.5, every transition for which both $\bar{s}_t$ and $\bar{s}_{t+1}$ are nonabsorbing satisfies*

$$e_{t+1} \leq \kappa_n e_t + L_{z^*} \Delta y_{\text{max}}, \quad (86)$$

and

$$\sup_{t: \bar{s}_t \neq \bar{s}_{\text{abs}}} \|z_t^{(n)} - z^*(x, y_t)\| \leq C_{\text{drift}}(n) := \kappa_n \max\left(E_0, \frac{L_{z^*} \Delta y_{\text{max}}}{1 - \kappa_n}\right) \leq \kappa_n E_0 + \frac{\kappa_n L_{z^*} \Delta y_{\text{max}}}{1 - \kappa_n}. \quad (87)$$

Lemma B.9 is the uniform specialization of Lemma B.4 with $F_t = T_{x, y_t}^{\circ n}$, $\xi_t = z^*(x, y_t)$, $\kappa_t = L_z^n$, and $d_t \leq L_{z^*} \Delta y_{\text{max}}$. The additional factor κ_n in $C_{\text{drift}}(n)$ appears because the value head reads the post-unroll latent.

Remark B.10 (When the drift bound is informative). The bound is useful only when its assumed constants are finite and calibrated on the policy-pair closure. Otherwise the direct residual certificate remains mathematically valid but uncalibrated, while the fixed-point comparison is vacuous.

Define

$$\bar{\varepsilon}_{\text{res},*} := \|\tilde{U}_* - \bar{\mathcal{T}}_K^{\bar{\pi}} \tilde{U}_*\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}}. \quad (88)$$

Table 1: Additional symbols and notations

Symbol	Meaning
$h, \bar{s} = (x, y, z, h)$	Remaining edit budget / clock-complete persistent state
$\mathcal{D}^\circ$	Declared nonabsorbing domain for a contraction invocation
$\mathcal{C}, \mathcal{C}^\circ$	Value domain / nonabsorbing latent domain
P_{rec}	Declared recurrent projection operator: Π_R for $R > 0$, or $\text{Id}_{\mathcal{Z}}$
$\mathcal{Z}_{\text{bd}}, V_{\text{max}}$	Declared bounded-head latent set / its uniform finite bound
$z^{(j)}(s), s \in \mathcal{C}^\circ$	Latent after finite depth j
U_j, U_*	Finite-depth value / optional fixed-point value, with scalar boundary
$B_{n,m}$	Finite path/discrepancy plus depth- m residual certificate
$\tilde{U}_n(\bar{s}), \tilde{U}_*(\bar{s})$	Persistent finite-depth value / memoryless reference
$\varepsilon_{\text{res},n}$	Episodic finite-depth Bellman residual
$\varepsilon_{\text{targ},q}$	Target-network conditional population residual at endpoint depth q
$R_{\text{targ},q}^P, R_{\text{targ},q}^{\text{sup}}$	Propagated-lag target bound / ordinary sup-lag fallback
$\bar{\varepsilon}_{\text{res},n}$	Persistent augmented-state finite-depth residual
$\bar{\varepsilon}_{\text{targ},q}$	Persistent target-network population residual at endpoint depth q
$\bar{R}_{\text{targ},q}^P, \bar{R}_{\text{targ},q}^{\text{sup}}$	Persistent propagated-lag target bound / sup-lag fallback
$\bar{\varepsilon}_{\text{res},*}$	Augmented residual of the fixed-point reference
$L_{z^*}, \Delta y_{\text{max}}, E_0$	Slow-drift parameters
$\varepsilon_A, \bar{\varepsilon}_A$	Episodic / augmented advantage error bounds
$\varepsilon_{A,\text{cand}}, \bar{\varepsilon}_{A,\text{cand}}$	Episodic / augmented candidate-defect suprema
$\varepsilon_{\text{res}}^*$	Architectural residual at U_*
g, Δ_g	Candidate expected advantage / span on the declared mixture or pair domain
$\hat{g}, \bar{\beta}_d, M_d$	Estimated candidate slope / one-sided signed-defect bound / certified slope
$\mathcal{C}_{\text{pair}}$	Common π/π_{cand} -invariant domain for step intervals
$c, b(s), \Xi_\alpha$	Signed centering and candidate estimator defects
$\tau(s), \delta_{\text{dep}}$	Policy overlap / deployment TV radius
ε_{CPI}	Uniform scalar bound on $ g $
$\kappa_n = L_z^n$	Contraction after n steps
$B_V(s)$	Baseline: $\mathbb{E}_{b \sim \pi}[\hat{Q}_V(s, b)]$

Then Lipschitzness of the head and Lemma B.9 give

$$\|\tilde{U}_n - \tilde{U}_*\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq L_V C_{\text{drift}}(n), \quad (89)$$

$$\|\tilde{U}_n - V^{\bar{\pi}}\|_{\infty, \bar{\pi}, \bar{\pi}_{\text{cand}}}^{\text{adv}} \leq L_V C_{\text{drift}}(n) + \frac{\bar{\varepsilon}_{\text{res},*}}{1 - \gamma^K}, \quad (90)$$

$$|\bar{\varepsilon}_{\text{res},n} - \bar{\varepsilon}_{\text{res},*}| \leq (1 + \gamma^K) L_V C_{\text{drift}}(n). \quad (91)$$

These are optional comparisons to a memoryless fixed-point reference; none is needed for Corollary B.2.

C NOTATION SUMMARY

We summarize notation here for convenience. Table 1 lists additional symbols used in proofs.

D CONSTRUCTIVE DEFINITION OF THE ADVANTAGE-EVALUATION CLOSURE

Definition D.1 (π -invariant Bellman domain). A measurable set $\mathcal{C}_\pi$ is a Bellman/value domain for π if $\rho(\mathcal{C}_\pi) = 1$ whenever the initial law is used and

$$P(\mathcal{C}_\pi \mid s, a) = 1 \quad \text{for } \pi(da \mid s)\text{-almost every } a \text{ and every } s \in \mathcal{C}_\pi.$$

Equivalently, the policy-induced kernel leaves $\mathcal{C}_\pi$ invariant. Theorem 6.1 needs only such a domain and does not need a candidate policy.

Definition D.2 (Advantage-evaluation domain). Fix current and candidate stochastic kernels π and π_{cand} . A measurable set $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ contains a π -invariant domain $\mathcal{C}_{\pi}$ and the common absorber and satisfies

1. for every $s \in \mathcal{C}_{\pi}$, $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \mid s, a) = 1$ for $\pi(da \mid s)$ - and $\pi_{\text{cand}}(da \mid s)$ -almost every action; and
2. for every $s \in \mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$, $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} \mid s, a) = 1$ for $\pi(da \mid s)$ -almost every action.

Thus the domain contains current-policy evaluation states, one current or candidate action, and every subsequent current-policy continuation. We write $\|f\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ for the ordinary supremum on this set, including the scalar absorbing boundary.

Definition D.3 (Mixture-occupancy and deployment domains). For a fixed α , a measurable $\mathcal{C}_{\text{mix}}$ is a mixture-occupancy domain if $\rho(\mathcal{C}_{\text{mix}}) = 1$ and it is invariant under both π and $\pi_{\alpha} = (1 - \alpha)\pi + \alpha\pi_{\text{cand}}$. It therefore supports the state laws at every time under the current policy and under repeated mixture decisions, not merely a one-deviation path. A deployment domain $\mathcal{D}_{\text{dep}}$ satisfies $\rho(\mathcal{D}_{\text{dep}}) = 1$ and is recursively invariant under both π_{α} and $\tilde{\pi}$. The CPI span theorem uses $\mathcal{C}_{\text{mix}}$; the deployment theorem uses $\mathcal{D}_{\text{dep}}$.

For a certificate simultaneous over more than one mixture weight, a measurable common policy-pair domain $\mathcal{C}_{\text{pair}}$ satisfies

$$\rho(\mathcal{C}_{\text{pair}}) = 1, \quad P_{\pi}(\mathcal{C}_{\text{pair}} \mid s) = P_{\pi_{\text{cand}}}(\mathcal{C}_{\text{pair}} \mid s) = 1 \quad \text{for every } s \in \mathcal{C}_{\text{pair}}. \quad (92)$$

By linearity of policy-induced kernels, every exact mixture π_{α} also leaves $\mathcal{C}_{\text{pair}}$ invariant. Thus $\mathcal{C}_{\text{pair}}$ is a valid single choice of $\mathcal{C}_{\text{mix}}$ for every $\alpha \in [0, 1]$.

Remark D.4 (Recovering candidate invariance from a positive mixture). Let $\alpha_0 > 0$. If a measurable set $\mathcal{E}$ is invariant under π and π_{α_0} , then for every $s \in \mathcal{E}$,

$$1 = P_{\pi_{\alpha_0}}(\mathcal{E} \mid s) = (1 - \alpha_0)P_{\pi}(\mathcal{E} \mid s) + \alpha_0 P_{\pi_{\text{cand}}}(\mathcal{E} \mid s),$$

and $P_{\pi}(\mathcal{E} \mid s) = 1$ forces $P_{\pi_{\text{cand}}}(\mathcal{E} \mid s) = 1$. This implication does not follow at $\alpha_0 = 0$ without separately assuming candidate invariance.

In finite or countable spaces these domains can be constructed with the usual positive-probability reachable sets and action supports. On arbitrary measurable action spaces we use the policy-almost-every-action clauses above; no topological action support is presumed.

Remark D.5 (Why the primary extension uses B_b). Theorem 6.1 works directly on $B_b(\mathcal{C})$ and needs no dominating measure. An alternative formulation on equivalence classes in a common $L^{\infty}(\nu)$ would additionally need initial and occupancy domination by ν , ν -nonsingularity of the relevant kernels, representative-independent Bellman integration, and the corresponding joint measurability conditions. Occupancy comparisons for general measures use total variation; vector ℓ_1 notation is reserved for finite or countable spaces.

E SHAPING CONVENTION AND ABSORBING-STATE NORMALIZATION

We use potential-based shaping with potential $\Phi(s) = c(x, y)$ for episodic $s = (x, y, h)$ and persistent $\bar{s} = (x, y, z, h)$ (the potential ignores the clock and latent), and $\Phi(s_{\text{abs}}) = C_{\text{max}}$ on their shared absorbing atom. We define

$$r(s, a, s') = r_0(s, a, s') + \gamma \Phi(s') - \Phi(s), \quad (93)$$

applying shaping on *all* transitions, including the absorbing self-loop.

For the absorbing self-loop, taking $r_0(s_{\text{abs}}, a, s_{\text{abs}}) = 0$ gives

$$r(s_{\text{abs}}, a, s_{\text{abs}}) = \gamma \Phi(s_{\text{abs}}) - \Phi(s_{\text{abs}}) = (\gamma - 1)C_{\text{max}}.$$

Thus every policy satisfies the fixed shaped boundary value

$$V^{\pi}(s_{\text{abs}}) = \sum_{t=0}^{\infty} \gamma^t (\gamma - 1)C_{\text{max}} = -C_{\text{max}}.$$

If one prefers the standard normalization $V(s_{\text{abs}}) = 0$, use the shifted potential $\tilde{\Phi}(s) := \Phi(s) - C_{\text{max}}$. This adds the constant reward $(1 - \gamma)C_{\text{max}}$ to every transition and shifts all returns by $+C_{\text{max}}$ independently of the policy. Accordingly, performance differences and optimal policies are unchanged.

F FINITE-HORIZON AND DEPLOYMENT-AWARE BOUNDS

This section uses the finite edit clock directly. Fix $0 \leq \gamma < 1$, an integer horizon $H \geq 0$, and one shared absorbing atom with boundary value

$$b := V_h^\pi(s_{\text{abs}}) = -C_{\text{max}} \quad \text{for every } h, \quad r_{\text{abs}} = (1 - \gamma)b.$$

Thus $b = r_{\text{abs}} + \gamma b$. Every reward interval below includes r_{abs} .

For each $h \in \{0, \dots, H\}$, let $\mathcal{C}_h$ be a measurable stage domain containing s_{abs} , with

$$\mathcal{C}_h^\circ := \mathcal{C}_h \setminus \{s_{\text{abs}}\}, \quad \mathcal{C}_0 = \{s_{\text{abs}}\}.$$

Value norms use $\|f\|_{\infty, \mathcal{C}_h} := \sup_{s \in \mathcal{C}_h} |f(s)|$; latent paths are defined only on $\mathcal{C}_h^\circ$.

Let $Q_h(dr, ds' | s, a)$ be the measurable controlled reward-transition kernel from stage h to stage $h-1$, let P_h be its state marginal, and let π_h be the current-policy stochastic kernel. Assume its induced kernel satisfies $P_h^\pi(\mathcal{C}_{h-1} | s) = 1$ for every $s \in \mathcal{C}_h$. If an initial law is used, assume $\rho(\mathcal{C}_H) = 1$. Assume the rewards and conditional expected rewards below are bounded and measurable. For $1 \leq \ell \leq h$ and $W \in B_b(\mathcal{C}_{h-\ell})$ with $W(s_{\text{abs}}) = b$, define

$$(\mathcal{T}_{h;\ell}^\pi W)(s) := \mathbb{E} \left[\sum_{i=0}^{\ell-1} \gamma^i r_i + \gamma^\ell W(S_\ell) \mid S_0 = s, A_i \sim \pi_{h-i}(\cdot | S_i) \right], \quad (94)$$

where an absorbed trajectory remains at the shared atom. Padding after early termination with absorbing self-loops and applying b at the end of the block account for the absorbing tail exactly once. These are current-policy Bellman/value domains; the repeated-mixture domains used for CPI are defined separately below.

Theorem F.1 (Measurable finite-horizon residual and finite-reference bound). *Fix $K \geq 1$. Let $U_{h,q} \in B_b(\mathcal{C}_h)$ for every $h \in \{0, \dots, H\}$ and every finite depth $q \geq 0$, and impose*

$$U_{h,q}(s_{\text{abs}}) = b, \quad U_{0,q} \equiv b \quad \text{for every finite } q \geq 0. \quad (95)$$

The true stage value satisfies $V_0^\pi \equiv b$ and $V_h^\pi(s_{\text{abs}}) = b$. For $h \geq 1$, set

$$\begin{aligned} \ell_h &:= \min\{K, h\}, & J_h &:= \left\lceil \frac{h}{K} \right\rceil, \\ \epsilon_{h,q} &:= \|U_{h,q} - \mathcal{T}_{h;\ell_h}^\pi U_{h-\ell_h,q}\|_{\infty, \mathcal{C}_h}, & D_{h;n,m} &:= \|U_{h,n} - U_{h,m}\|_{\infty, \mathcal{C}_h}, \end{aligned} \quad (96)$$

Then, for finite depths $0 \leq n < m$,

$$\|U_{h,n} - V_h^\pi\|_{\infty, \mathcal{C}_h} \leq D_{h;n,m} + \sum_{j=0}^{J_h-1} \gamma^{jK} \epsilon_{h-jK,m}. \quad (97)$$

Consequently, with $\epsilon_m^{(h)} := \max_{1 \leq r \leq h} \epsilon_{r,m}$,

$$\|U_{h,n} - V_h^\pi\|_{\infty, \mathcal{C}_h} \leq D_{h;n,m} + \frac{1 - \gamma^{KJ_h}}{1 - \gamma^K} \epsilon_m^{(h)}. \quad (98)$$

Taking the same finite depth as the reference gives

$$\|U_{h,q} - V_h^\pi\|_{\infty, \mathcal{C}_h} \leq \sum_{j=0}^{J_h-1} \gamma^{jK} \epsilon_{h-jK,q}. \quad (99)$$

For the recurrent specialization, fix one measurable frozen recurrent map F , a measurable initialization on each $\mathcal{C}_h^\circ$, a latent norm, and a bounded measurable scalar head V_ψ . Define only on $\mathcal{C}_h^\circ$

$$z_h^{(q+1)}(s) = F(z_h^{(q)}(s), s), \quad U_{h,q}(s) = V_\psi(z_h^{(q)}(s), x(s), y(s)).$$

For persistent state, $z_h^{(0)}(x, y, z, h) = z$ is the carried pre-unroll latent. An initialization that ignores h is an architectural specialization, not a premise of the residual argument. At the absorber use Eq. (95); no absorbing latent is defined. If the head is uniformly L_V -Lipschitz in the latent argument, then

$$D_{h;n,m} \leq L_V \sup_{s \in \mathcal{C}_h^\circ} \|z_h^{(n)}(s) - z_h^{(m)}(s)\| \quad (100)$$

$$\leq L_V \sup_{s \in \mathcal{C}_h^\circ} \sum_{q=n}^{m-1} \|z_h^{(q+1)}(s) - z_h^{(q)}(s)\| \quad (101)$$

$$\leq L_V \sum_{q=n}^{m-1} \sup_{s \in \mathcal{C}_h^\circ} \|z_h^{(q+1)}(s) - z_h^{(q)}(s)\|. \quad (102)$$

For $h = 0$, set $J_0 = 0$, $D_{0;n,m} = 0$, and interpret every residual and path sum as empty. If $\mathcal{C}_h^\circ$ is empty, the nonnegative latent suprema in the recurrent clause are defined to be zero.

Proof. For a fixed depth q , write $E_{h,q} := \|U_{h,q} - V_h^\pi\|_{\infty, \mathcal{C}_h}$ and $E_{0,q} = 0$. The stage Bellman recursion and the Markov-kernel expectation inequality give

$$\begin{aligned} E_{h,q} &\leq \epsilon_{h,q} + \|\mathcal{T}_{h:\ell_h}^\pi U_{h-\ell_h,q} - \mathcal{T}_{h:\ell_h}^\pi V_{h-\ell_h}^\pi\|_{\infty, \mathcal{C}_h} \\ &\leq \epsilon_{h,q} + \gamma^{\ell_h} E_{h-\ell_h,q}. \end{aligned}$$

On trajectories absorbed inside the block, the terminal-function difference is zero because both functions equal b . Unrolling visits stages $h, h-K, \dots, h-(J_h-1)K$; the last partial block ends at $E_{0,q} = 0$. This proves Eq. (99). Adding and subtracting $U_{h,m}$ proves Eq. (97); the uniform residual bound gives Eq. (98). Finally, head Lipschitzness gives Eq. (100); the triangle inequality for the pointwise telescoping path gives Eq. (101), and sup-sum interchange gives Eq. (102). $\square$

In a persistent-state application, the state is (x, y, z, h) . Whenever the current and candidate policies are analyzed in one augmented MDP, both use the same frozen recurrent map for action selection and for the carried post-unroll latent on each nonterminal transition.

For finite-horizon policy improvement, define

$$\eta_H(\pi) := \mathbb{E} \left[\sum_{t=0}^{H-1} \gamma^t r_t + \gamma^H b \right],$$

where early terminal trajectories are padded by the absorbing self-loop. Let d_t^π be the state law before decision t , including absorbed mass. For each h , let $\mathcal{M}_h$ be a measurable stagewise mixture-occupancy domain containing s_{abs} , with $\mathcal{M}_0 = \{s_{\text{abs}}\}$ and $\rho(\mathcal{M}_H) = 1$. Require the controlled successor kernel to satisfy

$$P_h(\mathcal{M}_{h-1} \mid s, a) = 1$$

for every $s \in \mathcal{M}_h$ and for $\pi_h(da \mid s)$ - and $\pi_{\text{cand},h}(da \mid s)$ -almost every action. This recursive invariance implies $d_{H-h}^\pi(\mathcal{M}_h) = d_{H-h}^{\pi_{\text{cand},h}}(\mathcal{M}_h) = 1$ and covers a candidate first step followed by every current-policy continuation. No topological action support is presumed. Define every stage advantage to be zero at the absorbing atom.

Theorem F.2 (Finite-horizon CPI with stagewise spans). *Fix bounded measurable stage rewards and the common terminal boundary b . For every $h \in \{1, \dots, H\}$, fix a current-policy kernel π_h , a candidate kernel $\pi_{\text{cand},h}$, and the exact pointwise probability-space mixture*

$$\pi_{\alpha,h}(\cdot \mid s) = (1 - \alpha)\pi_h(\cdot \mid s) + \alpha\pi_{\text{cand},h}(\cdot \mid s), \quad \alpha \in [0, 1]. \quad (103)$$

All policies act in the same fixed time-indexed MDP. Assume the relevant values, advantages, and estimator functions are bounded and measurable on $\mathcal{M}_h$. Define

$$g_h(s) := \mathbb{E}_{a \sim \pi_{\text{cand},h}(\cdot \mid s)}[A_h^\pi(s, a)], \quad \Delta_h := \sup_{s \in \mathcal{M}_h} g_h(s) - \inf_{s \in \mathcal{M}_h} g_h(s), \quad (104)$$

with $g_h(s_{\text{abs}}) = 0$. Let $\hat{A}_h$ be exactly centered:

$$\mathbb{E}_{a \sim \pi_h(\cdot \mid s)}[\hat{A}_h(s, a)] = 0 \quad \text{for every } s \in \mathcal{M}_h, \quad (105)$$

and define the signed candidate defect

$$\beta_h(s) := \mathbb{E}_{a \sim \pi_{\text{cand},h}(\cdot|s)} [\hat{A}_h(s, a) - A_h^\pi(s, a)]. \quad (106)$$

Set

$$\Xi_{H,\alpha} := \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^\pi} [\beta_{H-t}(s)]. \quad (107)$$

Define

$$\hat{L}_{H,\pi}(\pi_\alpha) := \eta_H(\pi) + \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{\substack{s \sim d_t^\pi, \\ a \sim \pi_{\alpha,H-t}(\cdot|s)}} [\hat{A}_{H-t}(s, a)]. \quad (108)$$

Then

$$\begin{aligned} \eta_H(\pi_\alpha) &\geq \hat{L}_{H,\pi}(\pi_\alpha) - \Xi_{H,\alpha} \\ &\quad - \alpha \sum_{t=0}^{H-1} \gamma^t \Delta_{H-t} [1 - (1 - \alpha)^t]. \end{aligned} \quad (109)$$

Because $\Xi_{H,\alpha}$ is signed, any established upper bound $B_{H,\alpha} \geq \Xi_{H,\alpha}$ may replace it. One valid choice is

$$B_{H,\alpha} := \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^\pi} |\beta_{H-t}(s)|. \quad (110)$$

Proof. The time-indexed performance-difference identity is

$$\eta_H(\pi_\alpha) - \eta_H(\pi) = \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^{\pi_\alpha}} [g_{H-t}(s)]. \quad (111)$$

The common terminal term cancels. Represent each mixture decision by an independent Bernoulli selector. Before decision t , the first t selectors have all chosen the current-policy component with probability $(1 - \alpha)^t$. Conditional on that event the state law is d_t^π , so for some probability law ν_t ,

$$d_t^{\pi_\alpha} = (1 - \alpha)^t d_t^\pi + [1 - (1 - \alpha)^t] \nu_t.$$

Therefore

$$\text{TV}(d_t^{\pi_\alpha}, d_t^\pi) \leq 1 - (1 - \alpha)^t. \quad (112)$$

The exponent is t , not $t + 1$, because d_t is measured before decision t . This selector proof needs no maximal-coupling kernel. Applying $|\mathbb{E}_\mu f - \mathbb{E}_\nu f| \leq \text{TV}(\mu, \nu) \text{span}(f)$ to g_{H-t} bounds the occupancy term. Exact centering gives

$$\hat{L}_{H,\pi}(\pi_\alpha) - \eta_H(\pi) = \alpha \sum_{t=0}^{H-1} \gamma^t \mathbb{E}_{s \sim d_t^\pi} [g_{H-t}(s) + \beta_{H-t}(s)].$$

Subtracting proves Eq. (109); moving absolute values inside the current-occupancy expectation proves Eq. (110). $\square$

Corollary F.3 (Uniform finite-horizon CPI specialization). *Under Theorem F.2, define*

$$\varepsilon_{\text{CPI},h} := \sup_{s \in \mathcal{M}_h} |g_h(s)|, \quad \varepsilon_{A,\text{cand},h} := \sup_{s \in \mathcal{M}_h} |\beta_h(s)|. \quad (113)$$

Then $\Delta_h \leq 2\varepsilon_{\text{CPI},h}$ and

$$\begin{aligned} \eta_H(\pi_\alpha) &\geq \hat{L}_{H,\pi}(\pi_\alpha) - \alpha \sum_{t=0}^{H-1} \gamma^t \varepsilon_{A,\text{cand},H-t} \\ &\quad - 2\alpha \sum_{t=0}^{H-1} \gamma^t \varepsilon_{\text{CPI},H-t} [1 - (1 - \alpha)^t]. \end{aligned} \quad (114)$$

For uniform bounds $\varepsilon_{A,\text{cand},h} \leq \varepsilon_{A,\text{cand}}$ and $\varepsilon_{\text{CPI},h} \leq \varepsilon_{\text{CPI}}$, define

$$G_H(q) := \sum_{t=0}^{H-1} q^t = \frac{1-q^H}{1-q}, \quad 0 \leq q < 1. \quad (115)$$

The total penalty in Eq. (114) is at most

$$\alpha \varepsilon_{A,\text{cand}} G_H(\gamma) + 2\alpha \varepsilon_{\text{CPI}} [G_H(\gamma) - G_H(\gamma(1-\alpha))]. \quad (116)$$

Corollary F.4 (Stagewise policy-overlap refinement). *Fix a finite evaluator depth q and bounded measurable approximations $U_{h,q}$ on the mixture domains. Write*

$$E_h := \|U_{h,q} - V_h^\pi\|_{\infty, \mathcal{M}_h}, \quad E_0 = 0,$$

where $U_{0,q} = V_0^\pi = b$ on $\mathcal{M}_0 = \{s_{\text{abs}}\}$. Define

$$\tau_h(s) := \text{TV}(\pi_{\text{cand},h}(\cdot | s), \pi_h(\cdot | s)), \quad \tau_{h,\infty} := \sup_{s \in \mathcal{M}_h} \tau_h(s).$$

Suppose $\hat{A}_h$ is the exactly centered one-step estimator derived from $U_{h-1,q}$, with reward and transition expectations evaluated exactly. Then

$$|\beta_h(s)| \leq 2\gamma E_{h-1} \tau_h(s), \quad (117)$$

$$\varepsilon_{A,\text{cand},h} \leq 2\gamma E_{h-1} \tau_{h,\infty}, \quad (118)$$

$$\mathbb{E}_{s \sim d_{H-h}^\pi} |\beta_h(s)| \leq 2\gamma E_{h-1} \mathbb{E}_{s \sim d_{H-h}^\pi} \tau_h(s). \quad (119)$$

In particular, the common terminal boundary gives $E_0 = 0$, so the stage-1 advantage-estimation defect is zero.

More generally, let $\ell_h = \min\{K, h\}$ and define the partial-block estimator

$$\hat{Q}_{h,q}^{(\ell_h)}(s, a) := \mathbb{E} \left[\sum_{i=0}^{\ell_h-1} \gamma^i r_i + \gamma^{\ell_h} U_{h-\ell_h,q}(S_{\ell_h}) \mid \begin{array}{l} S_0=s, A_0=a; \\ A_i \sim \pi_{h-i}(\cdot | S_i), 1 \leq i < \ell_h \end{array} \right], \quad (120)$$

and center it exactly under π_h . Its candidate defect satisfies

$$|\beta_h^{(\ell_h)}(s)| \leq 2\gamma^{\ell_h} E_{h-\ell_h} \tau_h(s), \quad (121)$$

$$\varepsilon_{A,\text{cand},h}^{(\ell_h)} \leq 2\gamma^{\ell_h} E_{h-\ell_h} \tau_{h,\infty}. \quad (122)$$

Thus a final partial block with $K > h$ has $\ell_h = h$ and zero bootstrap error because $E_0 = 0$.

If the residual theorem is used to certify E_h , its Bellman domains must contain the mixture domains $\mathcal{M}_h$ and satisfy the corresponding invariance assumptions. A one-deviation closure alone is insufficient. Under those enlarged-domain premises, take the estimator depth $q = n < m$. Equation (97) gives

$$\varepsilon_{A,\text{cand},h} \leq 2\gamma \tau_{h,\infty} \left[D_{h-1;n,m} + \sum_{j=0}^{J_{h-1}-1} \gamma^{jK} \epsilon_{h-1-jK,m} \right], \quad (123)$$

and Eq. (99) gives

$$\varepsilon_{A,\text{cand},h} \leq 2\gamma \tau_{h,\infty} \sum_{j=0}^{J_{h-1}-1} \gamma^{jK} \epsilon_{h-1-jK,n}. \quad (124)$$

For an ℓ_h -step estimator, the corresponding explicit bounds are

$$\varepsilon_{A,\text{cand},h}^{(\ell_h)} \leq 2\gamma^{\ell_h} \tau_{h,\infty} \left[D_{h-\ell_h;n,m} + \sum_{j=0}^{J_{h-\ell_h}-1} \gamma^{jK} \epsilon_{h-\ell_h-jK,m} \right], \quad (125)$$

$$\varepsilon_{A,\text{cand},h}^{(\ell_h)} \leq 2\gamma^{\ell_h} \tau_{h,\infty} \sum_{j=0}^{J_{h-\ell_h}-1} \gamma^{jK} \epsilon_{h-\ell_h-jK,n}. \quad (126)$$

The fallback follows from $\tau_h \leq 1$.

Proof. Let $e_{h-1} = U_{h-1,q} - V_{h-1}^\pi$ and $q_e(s, a) = \gamma \mathbb{E}[e_{h-1}(S_1) \mid s, a]$. Exact centering gives $\beta_h = \mathbb{E}_{\pi_{\text{cand},h}} q_e - \mathbb{E}_{\pi_h} q_e$. Recursive invariance gives a common action set of full current- and candidate-policy measure on which $q_e(s, a) \in [-\gamma E_{h-1}, \gamma E_{h-1}]$. The TV expectation inequality on this essential range proves Eqs. (117)–(119). For the partial block, the reward terms cancel and leave $q_e^{(\ell_h)}(s, a) = \gamma^{\ell_h} \mathbb{E}[e_{h-\ell_h}(S_{\ell_h}) \mid s, a]$, which lies in $[-\gamma^{\ell_h} E_{h-\ell_h}, \gamma^{\ell_h} E_{h-\ell_h}]$ on the analogous common essential action range. Substituting the two finite-horizon residual bounds proves the final displays. $\square$

For $H = 0$, $\eta_0(\pi) = b$, $G_0 = 0$, and every sum and penalty is empty. At $h = 0$, values equal the common boundary and no advantage is needed; set $g_0 = \Delta_0 = 0$ if convenient. When $K > h$, the last backup is the partial block $\ell_h = h$ and its terminal error vanishes. When $\gamma = 0$, only the $t = 0$ reward remains and the occupancy-shift penalty is zero. At $\alpha = 0$ every policy-shift term vanishes; at $\alpha = 1$ the selector bound is zero at $t = 0$ and one for $t \geq 1$. A nonzero common boundary cancels exactly in every policy comparison. In a folded zero-bootstrap representation, the terminal tail is already included in the folded reward and must not be added again.

Theorem F.5 (First-mismatch deployment perturbation). *Fix $0 \leq \gamma < 1$ and $\pi_\alpha = (1-\alpha)\pi + \alpha\pi_{\text{cand}}$ for $\alpha \in [0, 1]$. Let $\tilde{\pi}$ be a deployed stochastic kernel in the same fixed MDP as π_α , with the same initial law ρ . Assume the state and action spaces are standard Borel, and let $K(dr, ds' \mid s, a)$ be the measurable joint reward-transition kernel. The policy kernels then admit a measurable statewise maximal-coupling kernel.*

Let $\mathcal{D}_{\text{dep}}$ be measurable with $\rho(\mathcal{D}_{\text{dep}}) = 1$, and assume the policy-induced transition kernels of both π_α and $\tilde{\pi}$ leave $\mathcal{D}_{\text{dep}}$ invariant. For every $s \in \mathcal{D}_{\text{dep}}$, assume K assigns rewards in $[r_{\min}, r_{\max}]$ for $\pi_\alpha(da \mid s)$ - and $\tilde{\pi}(da \mid s)$ -almost every action. Write $\Delta_r := r_{\max} - r_{\min}$ and suppose

$$\sup_{s \in \mathcal{D}_{\text{dep}}} \text{TV}(\tilde{\pi}(\cdot \mid s), \pi_\alpha(\cdot \mid s)) \leq \delta_{\text{dep}}, \quad \delta_{\text{dep}} \in [0, 1]. \quad (127)$$

Then

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \delta_{\text{dep}}}{(1-\gamma)(1-\gamma + \gamma \delta_{\text{dep}})}. \quad (128)$$

The extra denominator factor satisfies $1 - \gamma + \gamma \delta_{\text{dep}} \geq 1 - \gamma$, so the first-mismatch constant improves the generic perturbation constant.

For an integer horizon $H \geq 0$, impose the same assumptions on the clock-complete deployment domain at every remaining horizon. If both returns use the same scalar terminal boundary b , then

$$\begin{aligned} |\eta_H(\tilde{\pi}) - \eta_H(\pi_\alpha)| &\leq \Delta_r \sum_{t=0}^{H-1} \gamma^t [1 - (1 - \delta_{\text{dep}})^{t+1}] \\ &= \Delta_r [G_H(\gamma) - (1 - \delta_{\text{dep}})G_H(\gamma(1 - \delta_{\text{dep}}))]. \end{aligned} \quad (129)$$

The common term $\gamma^H b$ cancels exactly, including when $b \neq 0$.

Proof. Couple the two processes from the same initial state. Whenever their states agree at decision t , use a measurable maximal coupling of $\pi_\alpha(\cdot \mid s_t)$ and $\tilde{\pi}(\cdot \mid s_t)$. Conditional on a matched history, their actions agree with probability at least $1 - \delta_{\text{dep}}$. When the actions agree, draw one shared reward and successor state from $K(\cdot \mid s_t, a_t)$. After the first action mismatch, use any measurable coupling.

Let E_t be the event that the actions agree at every decision $0, \dots, t$. Iterated conditioning gives

$$\Pr(E_t) \geq (1 - \delta_{\text{dep}})^{t+1}.$$

The exponent is $t + 1$ because the reward at time t follows decision t . On E_t , the two rewards at time t are identical. Since both rewards lie in an interval of width Δ_r ,

$$|\mathbb{E}[\tilde{r}_t] - \mathbb{E}[r_t^\alpha]| \leq \Delta_r \Pr(E_t^c) \leq \Delta_r [1 - (1 - \delta_{\text{dep}})^{t+1}].$$

Summing proves the finite-horizon bound; the common terminal scalar cancels. Letting $H \rightarrow \infty$ is valid because rewards are bounded, and

$$\begin{aligned} \sum_{t=0}^{\infty} \gamma^t [1 - (1 - \delta_{\text{dep}})^{t+1}] &= \frac{1}{1-\gamma} - \frac{1 - \delta_{\text{dep}}}{1 - \gamma(1 - \delta_{\text{dep}})} \\ &= \frac{\delta_{\text{dep}}}{(1-\gamma)(1-\gamma + \gamma \delta_{\text{dep}})}. \end{aligned}$$

1404
1405
1406
1407
1408
1409
1410
1411
1412
1413
1414
1415
1416
1417
1418
1419
1420
1421
1422
1423
1424
1425
1426
1427
1428
1429
1430
1431
1432
1433
1434
1435
1436
1437
1438
1439
1440
1441
1442
1443
1444
1445
1446
1447
1448
1449
1450
1451
1452
1453
1454
1455
1456
1457

□

Remark F.6 (Attainment by a first-mismatch process). The constants in Theorem F.5 are attained by a two-state MDP. Let ℓ be initial and u absorbing. At ℓ , action a_0 gives reward 0 and returns to ℓ , while a_1 gives reward Δ_r and enters u ; every action at u gives reward Δ_r and remains there. Let π_α choose a_0 at ℓ , and let $\tilde{\pi}$ choose a_1 with probability δ_{dep} and a_0 otherwise. Let the policies agree at u . The reward difference at time t is exactly $\Delta_r[1 - (1 - \delta_{\text{dep}})^{t+1}]$, so equality holds in both bounds, with terminal boundary $b = 0$ in the finite-horizon construction.

Corollary F.7 (Either-direction uniform-KL deployment bound). *Under Theorem F.5, suppose either*

$$\begin{aligned} \sup_{s \in \mathcal{D}_{\text{dep}}} D_{\text{KL}}(\tilde{\pi}(\cdot | s) \| \pi_\alpha(\cdot | s)) &\leq \kappa, \quad \text{or} \\ \sup_{s \in \mathcal{D}_{\text{dep}}} D_{\text{KL}}(\pi_\alpha(\cdot | s) \| \tilde{\pi}(\cdot | s)) &\leq \kappa \end{aligned}$$

for a finite κ . Define

$$\bar{\delta}_\kappa := \min \left\{ 1, \sqrt{\frac{\kappa}{2}} \right\}. \quad (130)$$

Then

$$|\eta(\tilde{\pi}) - \eta(\pi_\alpha)| \leq \frac{\Delta_r \bar{\delta}_\kappa}{(1 - \gamma)(1 - \gamma + \gamma \bar{\delta}_\kappa)}, \quad (131)$$

and

$$|\eta_H(\tilde{\pi}) - \eta_H(\pi_\alpha)| \leq \Delta_r [G_H(\gamma) - (1 - \bar{\delta}_\kappa)G_H(\gamma(1 - \bar{\delta}_\kappa))]. \quad (132)$$

Proof. Pinsker's inequality in either direction and $\text{TV} \leq 1$ show that the actual uniform TV discrepancy is at most $\bar{\delta}_\kappa$. Apply Theorem F.5 with $\delta_{\text{dep}} = \bar{\delta}_\kappa$; both right-hand sides are increasing in this radius. If the selected KL divergence is infinite, this corollary does not apply; the TV theorem remains available when Eq. (127) is established directly. □

Corollary F.8 (Occupancy-aware CPI with deployment discrepancy). *Assume Theorems 6.7 and F.5 hold in one fixed MDP with one initial law. Let $\mathcal{C}_{\text{mix}}$ satisfy Definition D.3, and let $\mathcal{D}_{\text{dep}}$ be measurable, carry the same initial law, and be recursively invariant under both π_α and $\tilde{\pi}$. Assume the standard-Borel, bounded-reward, and uniform-TV conditions of Theorem F.5. For any established one-sided bound $B_\alpha \geq \Xi_\alpha$,*

$$\begin{aligned} \eta(\tilde{\pi}) &\geq \hat{L}_\pi(\pi_\alpha) - \frac{B_\alpha}{1 - \gamma} - \frac{\gamma \alpha^2 \Delta_g}{(1 - \gamma)(1 - \gamma + \gamma \alpha)} \\ &\quad - \frac{\Delta_r \delta_{\text{dep}}}{(1 - \gamma)(1 - \gamma + \gamma \delta_{\text{dep}})}. \end{aligned} \quad (133)$$

A distilled, logit-interpolated, hidden-state-interpolated, parameter-interpolated, or otherwise realized policy is covered only when it acts in this same MDP and its TV or KL discrepancy from the exact probability-space mixture is established uniformly on $\mathcal{D}_{\text{dep}}$. For persistent state (x, y, z, h) , all policies must use the same frozen recurrent map that determines the carried-latent transition. Changing that map changes the MDP.

Proof. Lower-bound $\eta(\tilde{\pi})$ by $\eta(\pi_\alpha)$ minus Eq. (128), then apply Eq. (41) with $\Xi_\alpha \leq B_\alpha$. □

Remark F.9 (Boundary cases and uniform scope). At $\delta_{\text{dep}} = 0$, both deployment penalties are zero. At $\delta_{\text{dep}} = 1$, they are $\Delta_r/(1 - \gamma)$ and $\Delta_r G_H(\gamma)$. At $\gamma = 0$ and $H \geq 1$, both reduce to $\Delta_r \delta_{\text{dep}}$; at $H = 0$, the finite-horizon penalty is zero.

The residual, centering, TV, and KL premises are uniform on their declared domains. Replay-batch means, batch maxima, held-out finite maxima, and local finite-difference proxies are diagnostics. They do not establish the uniform premises.

G PROOF OF THE OCCUPANCY-AWARE CPI RESULT

The performance-difference identity and Bellman advantage identity used here are standard (Kakade & Langford, 2002). The exact selector calculation in Lemma 6.6 is specific to the pointwise probability mixture.

Proof of Theorem 6.7. For the true surrogate

$$L_\pi(\pi') := \eta(\pi) + \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\pi, a \sim \pi'(\cdot|s)} [A^\pi(s, a)],$$

linearity of the exact mixture and $\mathbb{E}_{a \sim \pi(\cdot|s)} A^\pi(s, a) = 0$ give

$$\begin{aligned} \hat{L}_\pi(\pi_\alpha) - L_\pi(\pi_\alpha) &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\pi} [(1-\alpha)c(s) + \alpha b(s)] \\ &= \frac{\Xi_\alpha}{1-\gamma}, \end{aligned}$$

which proves Eq. (40).

The performance-difference identity also gives

$$\eta(\pi_\alpha) - L_\pi(\pi_\alpha) = \frac{\alpha}{1-\gamma} (\mathbb{E}_{d_{\pi_\alpha}} g - \mathbb{E}_{d_\pi} g).$$

For any bounded measurable g and probability measures p, q , $|\mathbb{E}_p g - \mathbb{E}_q g| \leq \text{span}(g) \text{TV}(p, q)$. Applying this inequality on $\mathcal{C}_{\text{mix}}$ and then Lemma 6.6 yields

$$\eta(\pi_\alpha) - L_\pi(\pi_\alpha) \geq -\frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)}.$$

Combining the last display with the exact surrogate identity proves Eq. (41). Any $B_\alpha \geq \Xi_\alpha$ may replace the signed term. Under exact centering, $c = 0$ and $\Xi_\alpha = \alpha \mathbb{E}_{d_\pi} b \leq \alpha \varepsilon_{A, \text{cand}}$. Finally, $\Delta_g \leq 2 \sup_s |g(s)| = 2\varepsilon_{\text{CPI}}$, proving Eq. (45) and Eq. (62). $\square$

Corollary G.1 (Occupancy-averaged centering and candidate defects). *Under Theorem 6.7, let*

$$\varepsilon_{\text{cent}, d} := \mathbb{E}_{s \sim d_\pi} |c(s)|, \quad \varepsilon_{A, \text{cand}, d} := \mathbb{E}_{s \sim d_\pi} |b(s)|.$$

Then

$$\eta(\pi_\alpha) \geq \hat{L}_\pi(\pi_\alpha) - \frac{(1-\alpha)\varepsilon_{\text{cent}, d} + \alpha\varepsilon_{A, \text{cand}, d}}{1-\gamma} - \frac{\gamma\alpha^2\Delta_g}{(1-\gamma)(1-\gamma+\gamma\alpha)}. \quad (134)$$

Pointwise uniform bounds on $|c|$ and $|b|$ imply the corresponding looser version immediately. For a value-derived estimator, Eq. (32) supplies $\varepsilon_{A, \text{cand}, d} \leq 2\gamma\delta_V \mathbb{E}_{d_\pi} \tau(s)$.

Proof. The choice $B_\alpha = (1-\alpha)\varepsilon_{\text{cent}, d} + \alpha\varepsilon_{A, \text{cand}, d}$ upper-bounds Ξ_α . Substitute it in Eq. (41). $\square$

Proof of Corollary 6.8. The invariance of $\mathcal{C}_{\text{pair}}$ under both endpoint policies implies its invariance under every π_α . Hence Theorem 6.7 applies for each $\alpha \in [0, 1]$ with the same choice $\mathcal{C}_{\text{mix}} = \mathcal{C}_{\text{pair}}$, the same Δ_g , estimator, defect function, and one-sided defect bound. Exact centering removes the current-policy component:

$$\mathbb{E}_{s \sim d_\pi, a \sim \pi_\alpha(\cdot|s)} \hat{A}(s, a) = \alpha \hat{g}.$$

Together with the definition of the fixed candidate defect b , this gives the candidate-slope identity required by the certificate,

$$\hat{g} = \mathbb{E}_{s \sim d_\pi} [g(s) + b(s)]. \quad (135)$$

Consequently,

$$\mathbb{E}_{d_\pi} g = \hat{g} - \mathbb{E}_{d_\pi} b \geq \hat{g} - \bar{\beta}_d = M_d. \quad (136)$$

The performance-difference identity, the span-TV expectation inequality, and Lemma 6.6 now give

$$\eta(\pi_\alpha) - \eta(\pi) = \frac{\alpha}{1-\gamma} \mathbb{E}_{d_{\pi_\alpha}} g \geq \frac{\alpha}{1-\gamma} \left[\mathbb{E}_{d_\pi} g - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right] \geq \frac{\alpha}{1-\gamma} \left[M_d - \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha} \right].$$

This proves Eqs. (50) and (51). The exact signed choice meets Eq. (48) with equality, while $\mathbb{E}b \leq \mathbb{E}|b| \leq \varepsilon_{A,\text{cand}}$ proves the other two specializations. For

$$\varphi(\alpha) := \frac{\gamma\alpha\Delta_g}{1-\gamma+\gamma\alpha}, \quad \varphi'(\alpha) = \frac{\gamma\Delta_g(1-\gamma)}{(1-\gamma+\gamma\alpha)^2} \geq 0.$$

Thus $\varphi(0) = 0$ and $\max_{\alpha \in [0,1]} \varphi(\alpha) = \varphi(1) = \gamma\Delta_g$. If $\gamma > 0$ and $0 < M_d < \gamma\Delta_g$, then $M_d < \Delta_g$ because $\gamma < 1$, so $\Delta_g - M_d > 0$ and the threshold is positive. Moreover,

$$\frac{M_d(1-\gamma)}{\gamma(\Delta_g - M_d)} < 1 \iff M_d < \gamma\Delta_g,$$

which is exactly the branch premise. Rearrangement therefore gives Eq. (52) directly, with no outer minimum or further intersection with $[0, 1]$. The monotonicity and direct substitution give every stated boundary case. In particular, $\varphi(1) = \gamma\Delta_g$ places $M_d = \gamma\Delta_g$ in the all-step branch, and when $M_d = \Delta_g = 0$ the displayed lower bound is zero for every step. $\square$

G.1 PROOF OF COROLLARY 6.2

Proof. Because $\mathcal{T}_K^\pi$ is a γ^K -contraction with fixed point V^π ,

$$\begin{aligned} \|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} &\leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + \|\mathcal{T}_K^\pi U_n - \mathcal{T}_K^\pi V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \\ &\leq \varepsilon_{\text{res}, n} + \gamma^K \|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Rearranging gives (19). Lemma H.2 yields (20), and Corollary 6.4 sharpens the candidate bias to $2\gamma\tau_\infty\varepsilon_{\text{res}, n}/(1-\gamma^K)$. Substitution into Theorem 6.7 gives Eq. (21); $\tau_\infty \leq 1$ recovers the uniform fallback.

For the target-network forms, Proposition 5.2 and the same Bellman contraction argument give the following, where $\|\cdot\|_{\infty}^{\text{adv}}$ abbreviates $\|\cdot\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ within this proof:

$$(1-\gamma^K)\|U_q - V^\pi\|_{\infty}^{\text{adv}} \leq \|U_q - \mathcal{T}_K^\pi U_q\|_{\infty}^{\text{adv}} \leq R_{\text{targ}, q}^P,$$

which proves Eq. (22). Adding and subtracting U_m and applying this result at $q = m$ proves Eq. (23). For the stated exact value-derived estimator, Corollary 6.4 applies with $\delta_V \leq R_{\text{targ}, q}^P/(1-\gamma^K)$, proving Eq. (24). Substitution into the scalar specialization of Theorem 6.7 proves Eq. (25). Applying the same overlap result to the depth- n estimator with

$$\delta_V \leq \|U_n - U_m\|_{\infty}^{\text{adv}} + \frac{R_{\text{targ}, m}^P}{1-\gamma^K}$$

proves Eq. (26). No step identifies the target-network population operator with $\mathcal{T}_K^\pi$; the former only supplies the displayed upper bound on the latter's residual. $\square$

Remark G.2 (Self-bootstrap residual to fixed-reference residual). The uniform self-bootstrap residual below upper-bounds $\varepsilon_{\text{res}}^*$ up to the same truncation gap:

$$\varepsilon_{\text{res}}^* \leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + (1+\gamma^K)\|U_n - U_*\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \quad (137)$$

This follows by adding and subtracting U_n inside $U_* - \mathcal{T}_K^\pi U_*$ and using K -step Bellman contraction on the final term. It is distinct from the target-network population residual in Proposition 5.2; Eq. (4) controls the self-bootstrap residual only after also controlling the propagated target-network lag $\|P_\pi^K(\bar{V} - U_n)\|_{\infty}$; the ordinary sup-lag $\|\bar{V} - U_n\|_{\infty}$ is the nonexpansive fallback. Under Assumption 5.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the Lipschitz-head condition on that domain, combining (137) with Proposition A.2 gives

$$\varepsilon_{\text{res}}^* \leq \|U_n - \mathcal{T}_K^\pi U_n\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} + (1+\gamma^K) L_V \frac{L_z^n}{1-L_z} C_z. \quad (138)$$

Finite-batch or mean-square estimates of either residual remain diagnostics and do not establish these sup-norm inequalities.

H ADVANTAGE ESTIMATION AND BOUNDS

$$\hat{Q}_V(s, a) = \mathbb{E}_{s'}[r(s, a, s') + \gamma V(s')], \quad (139)$$

$$\hat{A}_V(s, a) = \hat{Q}_V(s, a) - \mathbb{E}_{b \sim \pi(\cdot | s)}[\hat{Q}_V(s, b)]. \quad (140)$$

Remark H.1 (Centering and baseline computation). The linear-in- α $\varepsilon_{A, \text{cand}}$ term in Theorem 6.7 relies on the *statewise centering* property

$$\mathbb{E}_{a \sim \pi(\cdot | s)}[\hat{A}(s, a)] = 0 \quad \text{for all } s.$$

This holds identically when $\hat{A}$ is defined as in (140) with the baseline computed exactly as $B_V(s) = \mathbb{E}_{b \sim \pi(\cdot | s)}[\hat{Q}_V(s, b)]$ (e.g., by summation over a finite action space).

If the baseline is approximated, centering may fail. We quantify this by the *centering defect*

$$\varepsilon_{\text{cent}} := \sup_{s \in \mathcal{C}_\pi} \left| \mathbb{E}_{a \sim \pi(\cdot | s)}[\hat{A}(s, a)] \right|.$$

The exact-centered scalar specialization in Eq. (45) corresponds to $\varepsilon_{\text{cent}} = 0$; the signed master inequality Eq. (41) also covers nonzero centering defects.

Lemma H.2 (Value to one-step advantage error). *Let $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ be the advantage-evaluation closure from Definition D.2, and let $\delta_V := \|V - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$. Then for every $s \in \mathcal{C}_\pi$ and for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a ,*

$$|\hat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V.$$

The factor 2γ is tight worst-case because error appears in both Q and the baseline.

Remark H.3 (Extensions to K -step and GAE-style advantages). Lemma H.2 is stated for the *one-step* estimator (140), hence the factor 2γ . If instead one defines a K -step action-value estimate

$$\hat{Q}_V^{(K)}(s, a) := \mathbb{E} \left[\sum_{t=0}^{K-1} \gamma^t r_t + \gamma^K V(s_K) \mid s_0 = s, a_0 = a, \pi \right],$$

and the corresponding centered advantage $\hat{A}_V^{(K)}(s, a) := \hat{Q}_V^{(K)}(s, a) - \mathbb{E}_{b \sim \pi(\cdot | s)}[\hat{Q}_V^{(K)}(s, b)]$ (with the baseline computed exactly), then the same argument yields

$$|\hat{A}_V^{(K)}(s, a) - A^\pi(s, a)| \leq 2\gamma^K \delta_V.$$

λ -return or GAE-style estimators lie outside the displayed bridge unless their estimator-specific uniform error is separately established. Such an error may be supplied directly as ε_A in Assumption 6.5; no GAE coefficient is derived here.

Corollary H.4 (Advantage error in terms of theorem parameters). *Under Theorem 6.1 and the exact-centering conditions of Lemma H.2,*

$$\varepsilon_A \leq 2\gamma B_{n, m}. \quad (141)$$

Under Assumption 5.1 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and the Lipschitz-head condition on that domain, taking $m \rightarrow \infty$ yields

$$\varepsilon_A \leq \underbrace{\frac{2\gamma \varepsilon_{\text{res}}^*}{1 - \gamma^K}}_{\text{architectural}} + \underbrace{2\gamma L_V \frac{L_z^n}{1 - L_z}}_{\text{unrolling}} C_z. \quad (142)$$

Here ε_A bounds the advantage error on $\mathcal{C}_\pi$ for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every action, and the intermediate value-error quantity in Lemma H.2 is taken over the advantage-evaluation closure $\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}}$ (Definition D.2).

I PROOFS

I.1 LEMMA I.1 (INTERIOR PROJECTION FIXED POINTS)

Lemma I.1 (Interior projection fixed points). *Let $\Pi_R : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be the Euclidean projection onto the closed ball $\{z : \|z\|_2 \leq R\}$ as in (54). Suppose $z^* = \Pi_R(u)$ for some $u \in \mathbb{R}^d$. If $\|z^*\|_2 < R$, then $z^* = u$ (i.e., the projection is inactive at u). Consequently, if z^* is a fixed point of $f_\theta(z, y, x) = \Pi_R(f_\theta(z, y, x))$ and $\|z^*\|_2 < R$, then $f_\theta(z^*, y, x) = z^*$ and Π_R is inactive at $f_\theta(z^*, y, x)$.*

Proof. By (54), if $\|u\|_2 \leq R$ then $\Pi_R(u) = u$. Since $z^* = \Pi_R(u)$ lies in the open ball $\|z^*\|_2 < R$, we have $z^* \neq R \frac{u}{\|u\|_2}$, so the second case of (54) cannot apply; hence $\|u\|_2 \leq R$ and $\Pi_R(u) = u = z^*$.

For the consequence: if $z^* = \tilde{f}_\theta(z^*, y, x) = \Pi_R(f_\theta(z^*, y, x))$ and $\|z^*\|_2 < R$, applying the first statement with $u = f_\theta(z^*, y, x)$ gives $f_\theta(z^*, y, x) = z^*$. $\square$

I.2 PROOF OF PROPOSITION A.2

Proof. Fix $s \in \mathcal{D}^\circ$. By Assumption 5.1, T_s maps $\mathcal{Z}_{\text{inv}}$ to itself (forward invariance) and is a contraction with modulus $L_z < 1$. Since $\mathcal{Z}_{\text{inv}}$ is a closed subset of the complete metric space $(\mathcal{Z}, \|\cdot\|)$, it is itself complete.

Banach's theorem yields a unique fixed point $z^*(s) \in \mathcal{Z}_{\text{inv}}$ and linear convergence (57).

Under the additional measurability premises in Eq. (59), the iterates are measurable by induction. Indeed, if $z^{(j)}$ is measurable, then $s \mapsto (s, z^{(j)}(s))$ is measurable into the product space, so $z^{(j+1)}(s) = T(s, z^{(j)}(s))$ is measurable. Banach convergence gives $z^{(j)}(s) \rightarrow z^*(s)$ pointwise. For every nonempty closed $F \subseteq \mathcal{Z}_{\text{inv}}$, continuity of distance and pointwise convergence give

$$d(z^*(s), F) = \lim_{j \rightarrow \infty} d(z^{(j)}(s), F).$$

The right side is a pointwise limit of measurable real functions, and

$$\{s : z^*(s) \in F\} = \{s : d(z^*(s), F) = 0\}.$$

The empty closed set has empty preimage. Since closed sets generate the Borel σ -algebra, z^* is measurable.

For the finite-unrolling bound, specialize to $\mathcal{D}^\circ = \mathcal{C}^\circ$ and take $z^{(0)}(s) = z_{\text{init}}(x(s), y(s))$:

$$z^*(s) - z^{(n)}(s) = \sum_{k=n}^{\infty} (z^{(k+1)}(s) - z^{(k)}(s)).$$

Contraction gives $\|z^{(k+1)}(s) - z^{(k)}(s)\| \leq L_z^k \|z^{(1)}(s) - z^{(0)}(s)\|$, so

$$\|z^*(s) - z^{(n)}(s)\| \leq \|z^{(1)}(s) - z^{(0)}(s)\| \sum_{k=n}^{\infty} L_z^k = \frac{L_z^n}{1 - L_z} \|z^{(1)}(s) - z^{(0)}(s)\|.$$

By definition (56), using the declared update $\tilde{f}_\theta$,

$$\|z^{(1)}(s) - z^{(0)}(s)\| = \|T_s(z_{\text{init}}(x(s), y(s))) - z_{\text{init}}(x(s), y(s))\| \leq C_z.$$

$\square$

I.3 PROOF OF LEMMA B.9

Proof. For $T_{x, y_t}^{\circ n}$ a κ_n -contraction (where $\kappa_n = L_z^n$):

$$\|z_{t+1} - z^*(x, y_t)\| = \|T_{x, y_t}^{\circ n}(z_t) - T_{x, y_t}^{\circ n}(z^*(x, y_t))\| \leq \kappa_n e_t.$$

By Assumption B.5:

$$\|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq L_{z^*} \|y_{t+1} - y_t\|_{\mathcal{E}} \leq L_{z^*} \Delta y_{\max}.$$

Thus:

$$e_{t+1} = \|z_{t+1} - z^*(x, y_{t+1})\| \leq \|z_{t+1} - z^*(x, y_t)\| + \|z^*(x, y_t) - z^*(x, y_{t+1})\| \leq \kappa_n e_t + L_{z^*} \Delta y_{\max}.$$

Unrolling this recurrence:

$$e_t \leq \kappa_n^t e_0 + L_{z^*} \Delta y_{\max} \sum_{j=0}^{t-1} \kappa_n^j = \kappa_n^t e_0 + \frac{L_{z^*} \Delta y_{\max} (1 - \kappa_n^t)}{1 - \kappa_n}.$$

Using $e_0 \leq E_0$ and noting that this expression is maximized either at $t = 0$ (giving E_0) or as $t \rightarrow \infty$ (giving $\frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}$), we have:

$$\sup_t e_t \leq \max \left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n} \right) \leq E_0 + \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n}.$$

For the used latent:

$$\|z_t^{(n)} - z^*(x, y_t)\| = \|T_{x, y_t}^{\circ n}(z_t) - z^*(x, y_t)\| = \|T_{x, y_t}^{\circ n}(z_t) - T_{x, y_t}^{\circ n}(z^*(x, y_t))\| \leq \kappa_n e_t,$$

where the second equality uses that $z^*(x, y_t)$ is the fixed point of T_{x, y_t} , hence also of $T_{x, y_t}^{\circ n}$, and the inequality follows from κ_n -contractivity of the n -fold composition. Thus:

$$\sup_t \|z_t^{(n)} - z^*(x, y_t)\| \leq \kappa_n \sup_t e_t \leq \kappa_n \max \left(E_0, \frac{L_{z^*} \Delta y_{\max}}{1 - \kappa_n} \right) = C_{\text{drift}}(n),$$

and the final “ $\leq$ ” in (87) gives the simpler additive upper bound. $\square$

I.4 PROOF OF THEOREM 6.1

Proof. Write $\|\cdot\|_{\infty}$ for the ordinary sup-norm on $\mathcal{C}$. Invariance and the Markov-kernel measurability property imply $P_{\pi}^K V \in B_b(\mathcal{C})$ whenever $V \in B_b(\mathcal{C})$; bounded r_K^{π} therefore makes $\mathcal{T}_K^{\pi}$ a self-map. For $V, W \in B_b(\mathcal{C})$,

$$\|\mathcal{T}_K^{\pi} V - \mathcal{T}_K^{\pi} W\|_{\infty} \leq \gamma^K \|V - W\|_{\infty}.$$

The bounded-reward premise makes the ordinary discounted policy value well defined, bounded, and measurable, with the uniformly convergent representation

$$V^{\pi} = \sum_{t=0}^{\infty} \gamma^t P_{\pi}^t r^{\pi}.$$

Splitting this series after K terms gives $V^{\pi} = \mathcal{T}_K^{\pi} V^{\pi}$. Completeness and Banach’s theorem then identify this policy value as the unique fixed point of $\mathcal{T}_K^{\pi}$. Thus

$$\begin{aligned} \|U_m - V^{\pi}\|_{\infty} &\leq \|U_m - \mathcal{T}_K^{\pi} U_m\|_{\infty} + \|\mathcal{T}_K^{\pi} U_m - \mathcal{T}_K^{\pi} V^{\pi}\|_{\infty} \\ &\leq \|U_m - \mathcal{T}_K^{\pi} U_m\|_{\infty} + \gamma^K \|U_m - V^{\pi}\|_{\infty}. \end{aligned}$$

Rearranging and then applying the triangle inequality to $U_n - U_m + U_m - V^{\pi}$ proves (8).

For every nonabsorbing s , Lipschitzness of the scalar head and the telescoping latent path give

$$\begin{aligned} |U_n(s) - U_m(s)| &\leq L_V \|z^{(n)}(s) - z^{(m)}(s)\| \\ &\leq L_V \sum_{j=n}^{m-1} \|z^{(j+1)}(s) - z^{(j)}(s)\|. \end{aligned}$$

The left side is zero at s_{abs} because its value is depth independent. Taking the supremum only over $\mathcal{C}^{\circ}$ for the latent endpoints proves Eq. (9). The triangle inequality for the telescoping latent path proves Eq. (10); using $\sup_s \sum_j g_j(s) \leq \sum_j \sup_s g_j(s)$ proves Eq. (11). No absorbing latent is defined.

Under the $\mathcal{D}^{\circ} = \mathcal{C}^{\circ}$ invocation of Assumption 5.1,

$$\|z^{(j+1)}(s) - z^{(j)}(s)\| \leq L_z^j \|z^{(1)}(s) - z^{(0)}(s)\| \leq L_z^j C_z.$$

Summing from $j = n$ through $m - 1$ yields (13). Proposition A.2 and the Lipschitz head give $\|U_m - U_*\|_{\infty} \rightarrow 0$. Moreover,

$$\|U_m - \mathcal{T}_K^{\pi} U_m\|_{\infty} - \|U_* - \mathcal{T}_K^{\pi} U_*\|_{\infty} \leq (1 + \gamma^K) \|U_m - U_*\|_{\infty} \rightarrow 0.$$

Taking $m \rightarrow \infty$ therefore proves (14). $\square$

Remark I.2 (Finite- R specialization of the truncation constant). Under the Appendix J initialization condition $\|z_{\text{init}}(x(s), y(s))\|_2 \leq R$ for every $s \in \mathcal{C}^\circ$, with $\mathcal{Z}_{\text{inv}} = \{z : \|z\|_2 \leq R\}$, the projection Π_R ensures $\|T_s(z_{\text{init}}(x(s), y(s)))\|_2 \leq R$ on the same domain. Therefore

$$C_z = \sup_{s \in \mathcal{C}^\circ} \|\tilde{f}_\theta(z_{\text{init}}(x(s), y(s)), y(s), x(s)) - z_{\text{init}}(x(s), y(s))\|_2 \leq 2R.$$

The contractive specialization of Theorem 6.1 therefore gives

$$\|U_n - V^\pi\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}} \leq \frac{\varepsilon_{\text{res}}^*}{1 - \gamma^K} + 2L_V \frac{L_z^n}{1 - L_z} R.$$

This makes R explicit only through the finite-unrolling term; it does not identify the dependence of $\varepsilon_{\text{res}}^*$ on R .

I.5 PROJECTION-ACTIVE THEORY

Projection-aware CPI penalty (full statement). Assume all premises of Proposition I.3 on $\mathcal{C}^\circ$, including saturation and the represented-initialization condition, together with the zero-anchor hypotheses of Proposition I.4, and suppose $\bar{L}_z^{\text{sat}}(R) < 1$. That proposition establishes Assumption 5.1 on $\mathcal{D}^\circ = \mathcal{C}^\circ$. Combining Eqs. (15) and (16) with $C_z \leq 2R$ yields a projection-aware version of (53):

$$\begin{aligned} \eta(\pi_{\text{new}}) &\geq \hat{L}_\pi(\pi_{\text{new}}) - \frac{2\alpha\gamma\tau_\infty}{1 - \gamma} \left(\frac{\varepsilon_0 + (1 + \gamma^K)L_V(R)R}{1 - \gamma^K} + 2L_V(R) \frac{(\bar{L}_z^{\text{sat}}(R))^n}{1 - \bar{L}_z^{\text{sat}}(R)} R \right) \\ &\quad - \frac{2\varepsilon_{\text{CPI}}\gamma\alpha^2}{(1 - \gamma)(1 - \gamma + \gamma\alpha)}. \end{aligned} \quad (143)$$

Projection affects both the architectural and truncation terms. The displayed bound does not make R a monotone performance dial because the residual, head Lipschitz factor, saturation margin, and optimizer trajectory may all change with the projection regime.

Proposition I.3 (Projection-induced contraction on a saturated annulus). Fix $R > 0$, let $B_R := \{z : \|z\|_2 \leq R\}$, and, for $s \in \mathcal{C}^\circ$, set $T_s(z) := \Pi_R(f_\theta(z, y(s), x(s)))$. Define the pre-projection domain-relative Lipschitz constant

$$L_z^{\text{pre}}(R) := \sup_{s \in \mathcal{C}^\circ} \sup_{\substack{z, z' \in B_R \\ z \neq z'}} \frac{\|f_\theta(z, y(s), x(s)) - f_\theta(z', y(s), x(s))\|_2}{\|z - z'\|_2}.$$

Assume there exists $\rho_R > R$ such that

$$\|f_\theta(z, y(s), x(s))\|_2 \geq \rho_R \quad \text{for all } s \in \mathcal{C}^\circ \text{ and } z \in B_R.$$

Then, for every $s \in \mathcal{C}^\circ$ and $z, z' \in B_R$,

$$\|T_s(z) - T_s(z')\|_2 \leq \bar{L}_z^{\text{sat}}(R) \|z - z'\|_2, \quad \bar{L}_z^{\text{sat}}(R) := \frac{R}{\rho_R} L_z^{\text{pre}}(R).$$

Projection also gives $T_s(B_R) \subseteq B_R$. Hence, if $\bar{L}_z^{\text{sat}}(R) < 1$ and $z_{\text{init}}(x(s), y(s)) \in B_R$ for every $s \in \mathcal{C}^\circ$, then Assumption 5.1 holds with $\mathcal{D}^\circ = \mathcal{C}^\circ$, $\mathcal{Z}_{\text{inv}} = B_R$, and $L_z \leq \bar{L}_z^{\text{sat}}(R)$. This conclusion does not assert contraction for input pairs not represented in $\mathcal{C}^\circ$.

Proof. For any u, v with $\|u\|_2, \|v\|_2 \geq \rho_R > R$,

$$\|\Pi_R(u) - \Pi_R(v)\|_2 = R \left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2.$$

Using

$$\left\| \frac{u}{\|u\|_2} - \frac{v}{\|v\|_2} \right\|_2^2 = \frac{\|u - v\|_2^2 - (\|u\|_2 - \|v\|_2)^2}{\|u\|_2 \|v\|_2} \leq \frac{\|u - v\|_2^2}{\rho_R^2},$$

we obtain

$$\|\Pi_R(u) - \Pi_R(v)\|_2 \leq \frac{R}{\rho_R} \|u - v\|_2.$$

Fix $s \in \mathcal{C}^\circ$ and apply this with $u = f_\theta(z, y(s), x(s))$ and $v = f_\theta(z', y(s), x(s))$ for $z, z' \in B_R$. Then

$$\|T_s(z) - T_s(z')\|_2 \leq \frac{R}{\rho_R} L_z^{\text{pre}}(R) \|z - z'\|_2. \quad \square$$

Proposition I.4 (Projection-aware residual via a state-dependent latent anchor). *Fix the discounted MDP, policy pair, γ , K , closure $\mathcal{C}$, and common absorber from Corollary 6.2. Require its bounded-function hypotheses, including $\mathcal{T}_K^\pi : B_b(\mathcal{C}) \rightarrow B_b(\mathcal{C})$. Write $\mathcal{C}^\circ = \mathcal{C} \setminus \{s_{\text{abs}}\}$. Assume Assumption 5.1 holds on $\mathcal{D}^\circ = \mathcal{C}^\circ$ with $\mathcal{Z}_{\text{inv}} = B_R$. Assume $(s, z) \mapsto T_s(z)$ is jointly measurable on $\mathcal{C}^\circ \times B_R$ and $s \mapsto z_{\text{init}}(x(s), y(s))$ is measurable into B_R . Proposition A.2 then makes the represented fixed-point map z^* measurable. Invoke Assumption A.3 with $\mathcal{D}^\circ = \mathcal{C}^\circ$ and $\mathcal{Z}_{\text{bd}} = B_R$, and require that the induced evaluator $U_*(s) = V_\psi(z^*(s), x(s), y(s))$ belongs to $B_b(\mathcal{C})$ with the common absorbing boundary. Fix any measurable anchor $z_{\text{anc}} : \mathcal{C}^\circ \rightarrow B_R$ and define*

$$\begin{aligned} B_{\text{anc}}(s) &:= V_\psi(z_{\text{anc}}(s), x(s), y(s)), & s \in \mathcal{C}^\circ, \\ B_{\text{anc}}(s_{\text{abs}}) &:= -C_{\text{max}}, \\ \varepsilon_{\text{anc}} &:= \|B_{\text{anc}} - \mathcal{T}_K^\pi B_{\text{anc}}\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Assume $B_{\text{anc}} \in B_b(\mathcal{C})$ and let

$$\begin{aligned} D_{\text{anc}}(R) &:= \sup_{s \in \mathcal{C}^\circ} \|z^*(s) - z_{\text{anc}}(s)\|_2, \\ L_V(R) &:= \sup_{s \in \mathcal{C}^\circ} \sup_{\substack{z, z' \in B_R \\ z \neq z'}} \frac{|V_\psi(z, x(s), y(s)) - V_\psi(z', x(s), y(s))|}{\|z - z'\|_2}. \end{aligned}$$

Then

$$\varepsilon_{\text{res}}^* \leq \varepsilon_{\text{anc}} + (1 + \gamma^K) L_V(R) D_{\text{anc}}(R). \quad (144)$$

In the special case $z_{\text{anc}}(s) \equiv 0$ on $\mathcal{C}^\circ$,

$$\begin{aligned} B_0(s) &:= V_\psi(0, x(s), y(s)), & s \in \mathcal{C}^\circ, \\ B_0(s_{\text{abs}}) &:= -C_{\text{max}}, \\ \varepsilon_{\text{res}}^* &\leq \varepsilon_0 + (1 + \gamma^K) L_V(R) R, \\ \varepsilon_0 &:= \|B_0 - \mathcal{T}_K^\pi B_0\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}. \end{aligned}$$

Proof. Add and subtract B_{anc} :

$$U_* - \mathcal{T}_K^\pi U_* = (B_{\text{anc}} - \mathcal{T}_K^\pi B_{\text{anc}}) + (U_* - B_{\text{anc}}) + (\mathcal{T}_K^\pi B_{\text{anc}} - \mathcal{T}_K^\pi U_*).$$

Taking $\|\cdot\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}$ and using the γ^K -contraction of $\mathcal{T}_K^\pi$ on the difference of its two images gives

$$\varepsilon_{\text{res}}^* \leq \varepsilon_{\text{anc}} + (1 + \gamma^K) \|U_* - B_{\text{anc}}\|_{\infty, \pi, \pi_{\text{cand}}}^{\text{adv}}.$$

For every $s \in \mathcal{C}^\circ$, both $z^*(s)$ and $z_{\text{anc}}(s)$ lie in B_R , so

$$\begin{aligned} |U_*(s) - B_{\text{anc}}(s)| &= |V_\psi(z^*(s), x(s), y(s)) - V_\psi(z_{\text{anc}}(s), x(s), y(s))| \\ &\leq L_V(R) \|z^*(s) - z_{\text{anc}}(s)\|_2 \leq L_V(R) D_{\text{anc}}(R). \end{aligned}$$

At the absorbing boundary, $U_*(s_{\text{abs}}) = B_{\text{anc}}(s_{\text{abs}}) = -C_{\text{max}}$, so the difference is zero. The zero-anchor specialization uses $D_{\text{anc}}(R) \leq R$. $\square$

I.6 PROOF OF LEMMA H.2

Proof. For any (s, a) ,

$$\widehat{Q}_V(s, a) - Q^\pi(s, a) = \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V(s') - V^\pi(s')].$$

If $s \in \mathcal{C}_\pi$, Definition D.2 gives $P(\mathcal{R}_{\pi, \pi_{\text{cand}}}^{\text{adv}} | s, a) = 1$ for $\pi(da | s)$ - and $\pi_{\text{cand}}(da | s)$ -almost every a . Thus $|V(s') - V^\pi(s')| \leq \delta_V$ for transition-almost every s' , and hence $|\widehat{Q}_V(s, a) - Q^\pi(s, a)| \leq \gamma \delta_V$.

Similarly,

$$|\mathbb{E}_{b \sim \pi(\cdot | s)} \widehat{Q}_V(s, b) - \mathbb{E}_{b \sim \pi(\cdot | s)} Q^\pi(s, b)| \leq \gamma \delta_V,$$

since $b \sim \pi(\cdot | s)$ and the same almost-sure closure condition applies. Subtracting these two bounds yields $|\widehat{A}_V(s, a) - A^\pi(s, a)| \leq 2\gamma \delta_V$. $\square$

I.7 PROOF OF COROLLARY B.2

Proof. The clock-complete state, frozen shared map F_n , and specified initial law define one augmented MDP for both action kernels. On its bounded measurable functions, $\bar{\mathcal{T}}_K^\pi$ is a γ^K -contraction in the augmented closure norm. In this proof, $\|\cdot\|_\infty^{\text{adv}}$ abbreviates $\|\cdot\|_{\infty, \bar{\pi}, \pi_{\text{cand}}}^{\text{adv}}$. Thus

$$\begin{aligned} \|\tilde{U}_n - V^\pi\|_\infty^{\text{adv}} &\leq \|\tilde{U}_n - \bar{\mathcal{T}}_K^\pi \tilde{U}_n\|_\infty^{\text{adv}} + \|\bar{\mathcal{T}}_K^\pi \tilde{U}_n - \bar{\mathcal{T}}_K^\pi V^\pi\|_\infty^{\text{adv}} \\ &\leq \bar{\varepsilon}_{\text{res}, n} + \gamma^K \|\tilde{U}_n - V^\pi\|_\infty^{\text{adv}}. \end{aligned}$$

Rearrangement proves (69). Lemma H.2, applied on the augmented closure with exact centering conditional on $\bar{s}$, gives (70). Corollary 6.4 supplies the $\bar{\tau}_\infty$ refinement. Under the exact pointwise mixture, substitution into Theorem 6.7 proves (73).

For the target-network statements, add and subtract its population backup on the same fixed augmented MDP:

$$\tilde{U}_q - \bar{\mathcal{T}}_K^\pi \tilde{U}_q = \left[\tilde{U}_q - (\bar{r}_K^\pi + \gamma^K P_K^\pi \bar{V}) \right] + \gamma^K P_K^\pi (\bar{V} - \tilde{U}_q).$$

The triangle inequality and nonexpansiveness of P_K^π prove Eq. (77). Bellman contraction then gives

$$(1 - \gamma^K) \|\tilde{U}_q - V^\pi\|_\infty^{\text{adv}} \leq \bar{R}_{\text{targ}, q}^P,$$

which proves Eq. (78). Adding and subtracting $\tilde{U}_m$ proves Eq. (79). The augmented version of Corollary 6.4, applied to the stated exact value-derived estimator, proves Eq. (81). Substitution into Theorem 6.7 proves Eq. (82); applying the same argument to the depth- n estimator and Eq. (79) proves Eq. (83). Throughout, q changes only an endpoint function, not F_n , the policies, or the augmented kernel. $\square$

For completeness, the optional slow-drift comparisons follow directly. Lemma B.9 and the Lipschitz head imply (89); adding and subtracting $\tilde{U}_*$ and applying the residual bound (5) proves (90). Finally, the reverse triangle inequality and Bellman contraction give

$$\begin{aligned} |\bar{\varepsilon}_{\text{res}, n} - \bar{\varepsilon}_{\text{res}, *}| &\leq \|(\tilde{U}_n - \tilde{U}_*) - (\bar{\mathcal{T}}_K^\pi \tilde{U}_n - \bar{\mathcal{T}}_K^\pi \tilde{U}_*)\|_\infty^{\text{adv}} \\ &\leq (1 + \gamma^K) \|\tilde{U}_n - \tilde{U}_*\|_\infty^{\text{adv}}, \end{aligned}$$

which proves (91).

J IMPLEMENTATION DETAILS

J.1 ALGORITHM PSEUDOCODE

Algorithm 2 presents the full UPI-TRM update step in persistent- z form with an explicit remaining-budget clock. For the episodic- z variant, use $s_k = (x, y_k, h_k)$, omit the carried latent from replay, and reinitialize $z_k \leftarrow z_{\text{init}}(x, y_k)$ at the start of each loop iteration. For a collected segment of length $\ell \leq K$, let d_{seg} indicate that its final transition terminates. The folded-reward implementation target is

$$G_{\text{fold}} = \sum_{k=0}^{\ell-1} \gamma^k r_k + \mathbf{1}\{-d_{\text{seg}}\} \gamma^\ell \bar{V}(\bar{s}_\ell). \quad (145)$$

Here $d_{\text{seg}} = 0$ implies $\ell = K$; early nonterminal segments are not admissible exact- K targets. On a terminal segment, the stored final reward is the formal terminal-transition reward plus γb , where $b = V^\pi(s_{\text{abs}})$. The zero padding and zero terminal bootstrap in Eq. (145) are therefore an accounting representation of the formal absorbing tail, not an additional self-loop. Under frozen-current-policy collection, this target has conditional population backup $r_K^\pi + \gamma^K P_K^\pi \bar{V}$ in the episodic model. Hence Proposition 5.2 converts its uniform population residual to the theorem’s self-bootstrap residual after adding the propagated lag $\gamma^K \|P_K^\pi (\bar{V} - U_n)\|_\infty$. Nonexpansiveness gives the ordinary sup-lag fallback $\gamma^K \|\bar{V} - U_n\|_\infty$. In the persistent model, replace every Bellman object by its augmented-MDP counterpart: in particular, $(\mathcal{E}, P_\pi, U_n, \mathcal{T}_K^\pi)$ becomes $(\bar{\mathcal{C}}, P_\pi, \tilde{U}_n, \bar{\mathcal{T}}_K^\pi)$, and the expected-reward term is the augmented K -step expected reward. The sharper lag term is then $\gamma^K \|P_K^\pi (\bar{V} - \tilde{U}_n)\|_\infty$,

Algorithm 2 UPI-TRM: one K -step collection and update

```

1: Input:  $x \in \mathcal{X}$ ,  $y_0 \in \mathcal{Y}$ , and  $h_0 \in \mathbb{N}_{\geq 1}$ ; for persistent- $z$  only,  $z_0 \in \mathcal{Z}$ 
2:  $\gamma \in [0, 1]$ ,  $n \in \mathbb{N}_0$ ,  $K \in \mathbb{N}_{\geq 1}$ ,  $\alpha \in [0, 1]$ , and target-retention coefficient  $\tau \in [0, 1]$ 
3: Recurrent mode: either  $R > 0$  with  $P_{\text{rec}} = \Pi_R$ , or projection disabled with  $P_{\text{rec}} = \text{Id}_{\mathcal{Z}}$ ; set  $\tilde{f}_\theta = P_{\text{rec}} \circ f_\theta$ 
4: Initialize  $\bar{s}_0 \leftarrow (x, y_0, z_0, h_0)$  and collected length  $\ell \leftarrow 0$ 
5: for  $k = 0$  to  $K - 1$  do
6:    $z_k^{(n)} \leftarrow \tilde{f}_\theta^{(n)}(z_k, y_k, x)$  // n inner steps
7:   Sample  $a_k \sim \pi_\phi(a \mid y_k, z_k^{(n)}, x)$ 
8:    $y_{k+1} \leftarrow \text{edit}(y_k, a_k; x)$ ; receive  $r_k$  and terminal flag  $d_k$ ; set  $h_{k+1} \leftarrow h_k - 1$ 
9:   if  $d_k$  or  $h_{k+1} = 0$  then
10:     $\bar{s}_{k+1} \leftarrow s_{\text{abs}}$ ; store  $(\bar{s}_k, a_k, r_k, s_{\text{abs}})$  in replay
11:     $\ell \leftarrow k + 1$ ; break
12:   else
13:     $z_{k+1} \leftarrow z_k^{(n)}$  // Carry only after a nonterminal transition
14:     $\bar{s}_{k+1} \leftarrow (x, y_{k+1}, z_{k+1}, h_{k+1})$ 
15:    Store  $(\bar{s}_k, a_k, r_k, \bar{s}_{k+1})$  in replay;  $\ell \leftarrow k + 1$ 
16:   end if
17: end for
18: Compute  $G_{\text{fold}}$  for the collected length  $\ell$  via (145)
19: Fit the value head  $\psi \rightarrow \psi^+$ ; the fixed-base protocol keeps the recurrent map  $\theta$  frozen
20: Form advantages  $\hat{A}$  via centered estimator (see below)
21: Policy-head step with  $F_n$  frozen  $\rightarrow$  candidate  $\pi_{\text{cand}}$ 
22:  $\pi_{\text{new}} \leftarrow (1 - \alpha)\pi_\phi + \alpha\pi_{\text{cand}}$ 
23: Deploy the exact pointwise mixture for the theorem; optional distillation or parameter interpolation is outside it
24:  $\bar{\psi}^+ \leftarrow \tau\bar{\psi} + (1 - \tau)\psi^+$  //  $\tau$  retains the old target
25: Output: fitted head  $\bar{\psi}^+$ , candidate  $\pi_{\text{cand}}$ , exact mixture  $\pi_{\text{new}}$ , and updated target head  $\bar{\psi}^+$ 

```

before the fallback $\gamma^K \|\bar{V} - \tilde{U}_n\|_\infty$. Empirical regression loss, population L^2 error, and a finite-batch maximum establish neither required sup norm.

The theorem-facing implementation seals the base policy and recurrent map F_n before collection, collects replay from that base, updates only the policy-independent value head, and optimizes one candidate policy head. The explicit mixture is evaluated or deployed without recursively promoting it to the next base. Thus repeated optimizer steps form one fixed-base proposal rather than an exact recursive CPI sequence.

The reference implementation stores the carried input and successor latents, the remaining-edit clock, terminal cause, terminal mask, and behavior log probability in replay. Recomputed episodic latents use the current evaluator, while persistent updates consume the recorded augmented state. On a terminal record, the environment has already folded the common absorbing tail into the reward, and every target path replaces the successor-value contribution by zero with a nonfinite-safe selection. The serialized terminal successor tensors are therefore never bootstrapped. This folded representation is return-equivalent to the single s_{abs} used in the proofs; it does not define puzzle-specific absorbing states.

Advantage estimation. The theory uses the centered estimator (140). The $\alpha \varepsilon_{A, \text{cand}} / (1 - \gamma)$ term in the readable specialization (45) requires *exact* statewise centering $\mathbb{E}_{a \sim \pi}[\hat{A}(s, a)] = 0$. This holds whenever the estimator is constructed to be exactly centered. For Eq. (140), the baseline $B_V(s) = \mathbb{E}_{b \sim \pi(\cdot|s)}[\hat{Q}_V(s, b)]$ must be evaluated exactly; the finite discrete implementation uses explicit summation. With a learned value baseline, as in standard GAE, centering is only approximate and must enter through the signed defect $c(s)$ or an upper bound B_α in Theorem 6.7. GAE is not covered by the displayed advantage bridge unless its uniform estimator error is supplied separately.

Operator-norm control and contraction-oriented details. The implementation can optionally apply operator-norm clamping on the inner $z \rightarrow z$ recursion (periodic rescaling using power-iteration estimates), followed by per-layer output scaling or an explicit scalar factor. The fixed-base exact path rejects scheduled clamping that would mutate the frozen recurrent map. These interventions do not certify the uniform modulus on the declared domain in Assumption 5.1. A possible local mini-batch

diagnostic is the finite-difference ratio

$$\hat{L}_z = \max_i \frac{\|f_\theta(z_i, y_i, x_i) - f_\theta(z'_i, y_i, x_i)\|}{\|z_i - z'_i\|}.$$

Enforcing forward-invariance. For an invocation on $\mathcal{D}^\circ$, the theorem needs $z_{\text{init}}(x(s), y(s)) \in \mathcal{Z}_{\text{inv}}$ and forward invariance only for represented $s \in \mathcal{D}^\circ$. A global projection-enabled implementation is one sufficient way to enforce these conditions. In projection-disabled mode, $P_{\text{rec}} = \text{Id}_{\mathcal{Z}}$ and no projection-induced invariant ball follows; R is absent rather than set to zero.

- With $R > 0$, use an initialization $z_{\text{init}}(x, y)$ with $\|z_{\text{init}}(x, y)\|_2 \leq R$, $\mathcal{Z}_{\text{inv}} = \{z : \|z\|_2 \leq R\}$. In the reference implementation, the initial latent comes from learned latent seeds (optionally augmented by an initialization encoder) and is projected into the same ball.
- After each update, apply Euclidean projection onto the ball: $z \leftarrow \Pi_R(z)$ with Π_R as in (54).
- Under the Euclidean norm this projection is 1-Lipschitz (non-expansive) and therefore cannot increase an independently established contraction modulus.
- Equivalently, define $\tilde{f}_\theta := \Pi_R \circ f_\theta$ and unroll $\tilde{f}_\theta^n$: $z^{(t+1)} = \Pi_R(f_\theta(z^{(t)}, y, x))$ for $t = 0, \dots, n-1$.

For persistent- z , one possible diagnostic is the fresh-initialization discrepancy $\hat{C}_{\text{fresh}}(n) = \max_t \|z_t^{(n)} - z_{\text{init}}(x, y_t)\|$. It measures separation from a fresh initialization; it is not a fixed-point tracking bound and is not a substitute for $C_{\text{drift}}(n)$ or the augmented residual.

Distillation. When distilling π_{new} to π_ϕ , we minimize:

$$\mathcal{L}_{\text{distill}}(\phi) = \mathbb{E}_{\bar{s} \sim \text{buffer}} [\text{KL}(\pi_{\text{new}}(\cdot | \bar{s}) \| \pi_\phi(\cdot | \bar{s}))].$$

This buffer-averaged objective is a heuristic diagnostic, not a uniform discrepancy certificate. The distilled policy is not the exact mixture. Corollary F.8 covers it only when a uniform TV or KL discrepancy is established on the deployment domain; a finite-batch $\text{KL}(\pi_{\text{new}} \| \pi_\phi)$ does not establish that premise. Parameter interpolation is also a different policy and must not be assigned a zero realization gap from the exact pointwise mixture.

K MAPPING TO ORIGINAL TRM ARCHITECTURE

We provide an explicit mapping between our formalism and the original TRM implementation (Jolicoeur-Martineau, 2025).

Our Formalism	TRM Implementation	Notes
$z \in \mathcal{Z}$	Hidden state	Latent reasoning state
$y \in \mathcal{Y}$	Output grid/sequence	Candidate solution
f_θ	Recurrent update	Single update step
n (inner steps)	Recursion depth	Fixed or adaptively chosen evaluation budget
$z_{\text{init}}(x, y)$	Initial hidden	Learned; projected when projection is enabled
$\pi_\phi(a y, z, x)$	Task head (reinterpreted)	TRM’s task head produces the refined answer; we reframe the answer update as a discrete edit policy
$\text{edit}(y, a; x)$	Continuous refinement	TRM: continuous y -update via another forward pass; we discretize into edit actions
Episodic- z	Not in original	Our simplification
Persistent- z	Original TRM	z carried across edits
Supervised objective	Training signal	Replaced by RL reward
Adaptive halting	Early stopping	Can be related to variable n

The key differences:

1. **Training signal:** The original TRM is trained with a supervised loss; we replace this with checker-based rewards and a CPI-style policy-improvement loop.

1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
2027
2028
2029
2030
2031
2032
2033
2034
2035
2036
2037
2038
2039
2040
2041
2042
2043
2044
2045
2046
2047
2048
2049
2050
2051

-
2. **Episodic- z vs. persistent- z :** We analyze both; the original TRM is persistent- z .
 3. **Halting:** The original TRM uses an adaptive halting mechanism; our analysis fixes n for tractability and treats n as an analysis parameter rather than a learned quantity.

L ETHICS AND BROADER IMPACT

Ethics statement. The paper studies a mathematical model of checker-trained Tiny Recursive Models on synthetic plan-editing tasks. We use no personal data, no human subjects, no crowdsourcing, and no deployed decision system. The main ethical risk is rhetorical: conditional bounds can be overstated as evidence about general reasoning or high-stakes automation. We report no empirical capability claim and state explicitly that the bounds do not certify improvement for a learned model.

Broader impact. The methodological contribution is to separate internal evaluator depth from outer policy improvement, which may help analyses of verifier-feedback systems with bounded internal compute. We make no deployment recommendation and keep the claims conditional on the displayed assumptions.